Persistent Information in Large Agent Populations: Scaling, Incentives, and Coordination

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Abstract

How do the scaling of information deposits and the incentives to produce them jointly determine the information environment that survives in a large population? We study one six-state population interacting through a decaying public board. With exogenously specified production, frequent deposits of size N^{-1} yield deterministic accumulation and ordinary tagged independence; rare order-one deposits survive as population-thinned common jumps and yield conditional tagged independence. A hybrid retains both. Direct coupling gives fixed-horizon quantitative bounds under explicit initialization assumptions. We then endogenize artifact production. Uniform nontransferable opportunities and readiness paid before assignment give each eligible player's effort marginal discovery intensity λ_M/N. A finite-population equilibrium exists and is unique within the bounded jointly continuous mild-HJB class. Without transfers, bounded private rewards force effort and aggregate artifact intensity to vanish at order N^{-1}. Yet in a seeded, zero-board, rare-only productive region, a specified coordinated policy yields an N-uniform positive per-capita welfare gain and retains common jumps with positive unconditional covariance of tagged task rewards. This is a fixed-policy comparison, not convergence of optimal controls or values. A two-state benchmark separates access allocation from capacity, classifies strategic productivity scalings, and permits balanced creator-credit implementation; a heterogeneous-state counterexample identifies its boundary. The results are conditional on the stated technology and institution, with no empirical validation for AI-agent systems.

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1 Scaling, incentives, and the shared environment

A discovery, file, or reusable procedure can remain available after its creation and change many users' later task outcomes. Whether a large population shares a nearly deterministic information stock or a consequential random history depends on two questions: how individual deposits scale, and who has an incentive to produce them. We ask how those two forces jointly determine the information environment that survives as the population grows.

The analysis follows one architecture throughout. Each agent has a binary task state and a communication state: unexposed, exposed, or adopter. Adopters can deposit information into a scalar board that decays and raises subsequent exposure rates. Deposits do not themselves change an agent's state. Frequent routine posts have small individual impact; rare artifacts have order-one impact. The same private transition rules operate in all regimes.

First hold production exogenous. An order-one per-adopter posting rate with board increment η/N becomes deterministic drift. A rate λ_M/N per adopter with increment h instead leaves a finite stream of common jumps. Acceptance of a common Poisson proposal depends on the current population, so the primitive noise is exogenous but the accepted intensity is population-generated. Running both channels gives a deterministic drift between random common jumps. Quantitative coupling and tagged-path results state exactly when these descriptions hold. The general positive-deposit scaling calculation concerns physical traffic, not equilibrium effort.

Next introduce incentives to produce the rare artifacts. Opportunities arrive at a fixed total rate, are allocated uniformly, cannot be reassigned, and require costly readiness before allocation. This gives an eligible individual marginal productivity λ_M/N at fixed readiness cost and fixed per-person reward. Bounded private continuation gains then suppress both individual effort and aggregate artifact intensity at order N^{-1}. An explicit continuous-board equilibrium construction makes this a result about a verified finite-population game, rather than an assumed effort rule.

Disappearance need not mean that artifacts are unproductive. In a seeded, zero-board setting without routine deposits, a fixed small coordinated effort raises average welfare by an amount bounded away from zero uniformly in N. That policy retains a common-jump limit and positive unconditional covariance of tagged task rewards. Artifacts improve outcomes in this region; the result does not attribute a welfare benefit to randomness itself. Fixed-policy convergence and the finite planner's existence do not interchange optimization with the population limit.

A solved two-state benchmark isolates the institution. Assigning all access to one creator changes the private problem even when total capacity is held fixed. A separate classification varies individual discovery productivity and includes linear costs, caps, and fixed positive discounting. Finally, balanced creator prizes implement that benchmark's planner under attribution and enforcement assumptions. The same price rule can fail with heterogeneous private states; a positive-duration deviation establishes the failure.

The contribution is this conditional connection between production, limiting information, and welfare. Common-noise limits, persistent environmental fields, and dynamic public-good incentives have substantial precedents, discussed together in Section 9. No novelty or empirical validity is inferred from assembling them. The board is deliberately coarse: it omits content, permissions, artifact-specific exposure, and spatial structure. Productivity is known, not learned; endogenous hierarchy and efficient specialization are not results of this model. Shared files and reusable procedures are hypothetical motivations, not an identified empirical setting.

Section 2 gives the common model and initialization assumptions. Section 3 develops the exogenous limits. The access benchmark and finite game follow in Sections 4–5. Sections 6–7 connect productive coordination to the equilibrium and fixed-policy limits; Section 8 examines creator credit. Numerical evidence and limitations close the paper. Appendix A gives the common probability

estimates, Appendix B gives potential verification and numerical formulas, and Appendix C retains the exposure/adoption threshold as a secondary application of the weak limit.

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2 One finite-population architecture

2.1 Individual and empirical states

Let

$$S = \{0,1\} \times \{\mathsf{U}, \mathsf{E}, \mathsf{A}\}, \qquad |S| = 6,$$

and write $Y_i^N = (X_i^N, C_i^N) \in S$ for agent i. The empirical law and adopter share are

$$\mu_t^N = \frac{1}{N} \sum_{i=1}^{N} \delta_{Y_i^N(t)}, \qquad a(\mu) = \mu(\{(0,\mathsf{A}), (1,\mathsf{A})\}).$$

The persistent information state is $B_t^N \in \mathbb{R}_+$. The type space is a singleton. Permanent types are not hidden in the notation; adding them would require type-indexed empirical masses and type-specific tagged limits.

Put $c_{\mathsf{U}} = c_{\mathsf{E}} = c_0$ and $c_{\mathsf{A}} = c_1$. Conditional on the current state, individual transitions have rates

$$\begin{array}{lll} (0,c) \to (1,c) & \text{at rate } c_c, & (1,c) \to (0,c) \text{ at rate } r, \\ (x,\mathsf{U}) \to (x,\mathsf{E}) & \text{at rate } \rho B^N, & (x,\mathsf{E}) \to (x,\mathsf{A}) \text{ at rate } \alpha, \\ (x,\mathsf{E}) \to (x,\mathsf{U}) & \text{at rate } \nu, & (x,\mathsf{A}) \to (x,\mathsf{U}) \text{ at rate } \delta. \end{array}$$

All parameters are finite and nonnegative. Exposure is the only unbounded rate, and it grows linearly with the board.

2.2 Exogenously specified information channels

In the frozen model, production intensities below are prescribed; no player chooses them and no payoff maximization is asserted. Section 5 extends this architecture by replacing the artifact rate with an effort-dependent one. Frozen unit production is an admissible game policy only if $\bar{u} \geq 1$; the probability model itself imposes no readiness cap.

An adopter is eligible for two information-event families. Neither event changes the triggering agent's individual state.

Table 1: One architecture with two fixed information channels.

family	per-adopter rate	board jump	weak	rare	hybrid
routine post	λ_w	η/N	on	off	on
major artifact	λ_M/N	h	off	on	on

Between jumps, $dB_t^N = -\kappa B_t^N dt$. Thus routine posts have aggregate rate $N\lambda_w a(\mu^N)$, whereas major artifacts have aggregate rate $\lambda_M a(\mu^N)$. The channels are semantically distinct fixed event

families. The labels weak and rare refer to their rate/jump scalings, not to different private-agent state transitions.

For $y, y' \in S$, let $v_{yy'} = \delta_{y'} - \delta_y$, and let $q_{yy'}(b)$ denote the individual rates above. If $w, m \in \{0, 1\}$ indicate whether the weak and major channels are active, the generator on smooth $f(\mu, b)$ is

$$\begin{aligned}\mathcal{L}_N^{w,m} f(\mu, b) = & -\kappa b\, \partial_b f(\mu, b) \\ & + N \sum_{y,y'} \mu(y) q_{yy'}(b) \left[f\left(\mu + \frac{v_{yy'}}{N}, b\right) - f(\mu, b) \right] \\ & + wN\lambda_w a(\mu) \left[f\left(\mu, b + \frac{\eta}{N}\right) - f(\mu, b) \right] \\ & + m\lambda_M a(\mu) \left[f(\mu, b + h) - f(\mu, b) \right]. \end{aligned} \tag{1}$$

The choices $(w, m) = (1, 0), (0, 1), (1, 1)$ give the weak, rare, and hybrid models. The artifact-disabled baseline also allows $(w, m) = (0, 0)$.

For major events, N individual clocks of rate $\lambda_M \mathbf{1}_{\{C_i^N = \mathsf{A}\}}/N$ can be superposed exactly. Let $\Pi^0(dt, du)$ be a Poisson random measure (PRM) on $\mathbb{R}_+^2$ with intensity $dt\, du$. The accepted count is

$$J_t^N = \int_{(0,t]\times\mathbb{R}_+} \mathbf{1}_{\{u \le \lambda_M a(\mu_{s-}^N)\}} \Pi^0(ds, du). \tag{2}$$

The primitive proposal measure is exogenous. The accepted process is endogenous in the precise sense that its predictable intensity is determined by the current empirical adopter share.

2.3 Well-posedness and standing assumptions

On a fixed horizon $[0, T]$, the board admits the joint domination

$$\sup_{t \le T} B_t^N \le B_0^N + w\frac{\eta}{N} P_w(N\lambda_w T) + hP_M(L_0 T), \tag{3}$$

for independent dominating Poisson counts. Here $L_0 = m\lambda_M$ for frozen production, and $L_0 = \lambda_M \bar{u}$ for the bounded controls introduced below. The same bound holds for every admissible predictable control because its artifact intensity never exceeds L_0. For bounded initial boards it supplies uniform polynomial moments and the exponential moments used below. Conditional on this bound, total individual transition intensity is finite. The competing-clock construction is therefore nonexplosive and unique in law on finite horizons.

For quantitative theorems, fix $p > 2$ and assume

$$0 \le B_0^N \le K \quad \text{a.s.}, \qquad \varepsilon_{N,p} := \left\{ \mathbb{E}\left[(\|\mu_0^N - \mu_0\|_1 + |B_0^N - b_0|)^p \right] \right\}^{1/p} \longrightarrow 0, \tag{4}$$

where $(\mu_0, b_0) \in \mathcal{P}(S) \times \mathbb{R}_+$ is deterministic and initial data are independent of future driving PRMs. Only deterministic limiting initial data are covered by the displayed quantitative and tagged statements. With additional conditional initialization hypotheses, common random initial data would instead be included in the common sigma-field. Constants denoted C_T may depend on T, p, K and the fixed model parameters, but never on N.

We use the norm $\|(\mu, b)\| = \|\mu\|_1 + |b|$. All process-level statements are on $\mathcal{D}([0, T], \mathcal{P}(S) \times \mathbb{R}_+)$, or the corresponding product including $\mathbb{N}$, with Skorokhod J_1 topology. Quantitative couplings use the stronger uniform metric. For tagged paths, the state space is $\mathcal{D}([0, T], S)$ with J_1.

We regard probability vectors as rows and write $Q(b) = (q_{yy'}(b))$, including diagonal entries $q_{yy}(b) = -\sum_{y' \neq y} q_{yy'}(b)$. Thus

$$F(\mu, b) = \mu Q(b) = \sum_{y,y'} v_{yy'} \mu(y) q_{yy'}(b).$$

The notation c_0, c_1 denotes task-completion rates; $c(u)$ below denotes readiness cost. Routine productivity is always λ_w, rare productivity is λ_M, and λ is reserved for the separate two-state benchmark.

3 Exogenous deposition and limiting environments

3.1 Physical deposition scaling

The generator makes the first-order contrast transparent. At adopter share $a^N = a(\mu^N)$, the predictable board drift and quadratic-variation densities from the two channels are

	drift	quadratic variation
weak	$\eta \lambda_w a^N$	$\eta^2 \lambda_w a^N / N$
major	$h \lambda_M a^N$	$h^2 \lambda_M a^N$.

The weak martingale vanishes while its compensator remains. The major martingale and its jumps remain of order one. In the hybrid model the two rows add, so only the weak fluctuation disappears.

This calculation extends to a positive-deposit family whose per-agent event rate is $\lambda_0 N^{-\zeta}$ and whose board impact is $\eta_0 N^{-\xi}$, with fixed $\lambda_0, \eta_0 > 0$. It is useful only after specifying that the eligible population share stays of order one.

Proposition 3.1 (Uncompensated scaling classification)**.** *For the positive, uncompensated information process just described, the aggregate rate, first-order drift, and predictable quadratic variation have orders*

$$N^{1-\zeta}, \qquad N^{1-\zeta-\xi}, \qquad N^{1-\zeta-2\xi},$$

respectively. In the quadrant $\zeta, \xi \geq 0$, order-one positive uncentered drift requires $\zeta + \xi = 1$. On this line, $\xi > 0$ gives vanishing jumps and quadratic variation of order $N^{-\xi}$, while $(\zeta, \xi) = (1, 0)$ gives a finite-rate, common-jump endpoint with nonvanishing jumps and state-dependent acceptance. If $\zeta + \xi > 1$ the channel vanishes at first order; if $\zeta + \xi < 1$ its positive drift diverges absent saturation or rescaling.

The weak model is $(\zeta, \xi) = (0, 1)$ and the rare model is $(1, 0)$. The line $\zeta + 2\xi = 1$ merely identifies order-one quadratic variation. When $\xi > 0$ on that line, the uncompensated drift grows as N^{ξ}. A Brownian limit would require centering, diverging event counts, and a separate Lindeberg or martingale-CLT argument. No diffusion theorem is asserted here.

These are order calculations conditional on an order-one eligible share; feedback that drives eligibility to zero can change them. At the finite-jump endpoint, the accepted count is not generally a Poisson process with a deterministic rate: its intensity depends on the current population. A compound-Poisson description requires a separately fixed acceptance rate. The strategic productivity exponent in Proposition 4.3 changes incentives and is a different classification.

Table 2: Finite channels and their large-population consequences.

regime	finite information traffic	limiting environment	tagged paths
weak	rate $N\lambda_w a^N$, jump η/N	deterministic drift $\eta\lambda_w a(\mu_t)dt$; board martingale vanishes	ordinary product law because the limiting board is deterministic
rare	rate $\lambda_M a^N$, jump h	state-thinned common count J; board has jumps $h\,dJ_t$	product only conditional on the common proposal measure/environment
hybrid	both independent channels	weak drift plus the rare common-jump count	conditional product law; weak traffic adds no limiting common randomness

3.2 Comparison of the three limits

Table 2 separates four objects that are easy to conflate: the finite information channel, the limiting board, the status of the empirical law, and the appropriate tagged-particle product statement.

The deterministic label in the weak row applies to the limit, not to the finite system. For each N, routine posting and individual state changes are random. Their martingales have order $N^{-1/2}$ on a fixed horizon, so the random paths concentrate around the deterministic coupled ODE. A nonlinear feedback through b can amplify or damp perturbations, but it does not restore order-one board noise under the fixed-horizon assumptions of the theorem.

The rare row is different before any limiting argument: there are already only order-one major events in an order-one time interval. Their triggering agents need not be retained in the state because the frozen model assigns no state change to the trigger. Superposition therefore loses no information relevant to future dynamics. In models with attribution states or trigger-dependent marks, discarding identity would no longer be harmless; those models are outside the present claim.

The limit measure μ_t in the rare and hybrid rows is random but continuous. It is the conditional time marginal of a nonlinear chain in the common board environment. The common board jumps, while each individual path changes only through its own transition clocks. Thus randomness in a measure-valued limit does not require a jump of the measure itself. It is enough that a common event changes every agent's subsequent transition rates.

3.3 Convergence language

The quantitative estimates below use a shared construction and the uniform distance on $[0, T]$. This is stronger than the reported J_1 convergence. The strength is natural here: weak limits are continuous, and in rare/hybrid modes finite and limiting accepted jumps occur at exactly the same proposal times unless a thinning mismatch occurs. The probability and expected consequences of such a mismatch are controlled directly.

Ordinary propagation of chaos is a statement about an unconditional, fixed-k path law converging to a deterministic product. Conditional propagation of chaos is a statement about random conditional laws. Formula (9) below uses L^1 convergence of conditional expectations of path-product tests. Integrating out the common PRM generally produces a mixture of products, not a product. The difference is structural, not terminological: observing or fixing the common artifact history changes the dependence question being asked.

3.4 Weak-information limit

The weak limiting equations are

$$\dot{\mu}_t = F(\mu_t, b_t), \qquad \dot{b}_t = -\kappa b_t + w\eta\lambda_w a(\mu_t). \tag{5}$$

They preserve the probability simplex and nonnegative board, and they have a unique global solution on every finite horizon.

Theorem 3.2 (Weak-information limit and ordinary chaos)**.** *Assume* (4) *and disable artifacts* ($m = 0$)*, with* $w = 0$ *or* 1*. Then for every* $T < \infty$*,*

$$\mathbb{E}\sup_{t\le T}\left(\|\mu_t^N - \mu_t\|_1 + |B_t^N - b_t|\right) \le C_T\left(\varepsilon_{N,p} + N^{-1/2}\right). \tag{6}$$

Consequently (μ^N, B^N) *converges in probability in the uniform metric, and hence in* J_1*, to the deterministic solution of* (5)*.*

If the initial array is exchangeable and μ_0*-chaotic, then for every fixed* k*,*

$$(Y_1^N, \dots, Y_k^N) \Longrightarrow (\bar{Y}_1, \dots, \bar{Y}_k) \quad \text{in } \mathcal{D}([0,T], S)^k,$$

where the $\bar{Y}_i$ *are independent time-inhomogeneous chains with rates* $q_{yy'}(b_t)$ *and common path law* $\mathcal{P}_\mu$*. Thus the limiting law is* $\mathcal{P}_\mu^{\otimes k}$ *and the time-t marginal of each chain is* μ_t*.*

The initialization statement contains two distinct requirements. The quantitative population estimate uses (4). For tagged convergence, exchangeable chaos supplies couplings of any fixed number of initial tags with i.i.d. μ_0 samples; in the finite state space it also implies convergence of the empirical population vector in every finite L^p. Board convergence is a separate part of (4). No Wasserstein rate uniform in the state-space dimension is claimed.

Ordinary chaos is possible here because the board limit is deterministic. The agents communicate through the finite system, but the fluctuations of their shared weak-information stock disappear under the normalization. The result is therefore not the claim that communication leaves agents independent at finite N.

Proofs, including the artifact-disabled case $w = 0$, are in Appendix A.

3.5 Rare artifacts and the hybrid limit

On the same primitive PRM as (2), define the rare limiting system

$$\begin{aligned}
\dot{\mu}_t &= F(\mu_t, B_t),\\
B_t &= b_0 - \int_0^t \kappa B_s\, ds + hJ_t,\\
J_t &= \int_{(0,t]\times\mathbb{R}_+} \mathbf{1}_{\{u\le\lambda_M a(\mu_{s-})\}}\Pi^0(ds, du).
\end{aligned} \tag{7}$$

Only points with $u \le \lambda_M$ can be accepted, so there are finitely many proposals on a finite horizon. Recursively solving the ODE between proposal times gives a pathwise unique strong solution measurable in Π^0. The population law μ is continuous; B and J are càdlàg, and a board jump changes all later exposure intensities.

Let

$$\mathcal{F}_T^0 = \sigma\Big(\Pi^0((0,t]\times D) : t \le T, D \subset [0,\lambda_M] \text{ Borel}\Big)$$

with the usual augmentation. The limiting initial data here are deterministic, as in (4). Let $\mathcal{P}^0$ be the conditional path law on $\mathcal{D}([0,T], S)$ of a chain with rates $q_{yy'}(B_t)$ driven by an independent idiosyncratic PRM. Its conditional time marginal is μ_t.

Theorem 3.3 (Rare-artifact limit and conditional chaos)**.** *Assume* (4) *and use rare mode. Couple the finite and limiting systems through the same* Π^0*. Then*

$$\mathbb{E}\sup_{t\le T}\left(\|\mu_t^N-\mu_t\|_1+|B_t^N-B_t|+|J_t^N-J_t|\right)\le C_T\left(\varepsilon_{N,p}+N^{-1/2}\right). \tag{8}$$

Hence the triple converges in probability in the uniform metric and in $\mathcal{D}([0,T],\mathcal{P}(S)\times\mathbb{R}_+\times\mathbb{N})$ *with product* J_1 *topology.*

Suppose also that the initial array is exchangeable and conditionally μ_0*-chaotic, independently of future idiosyncratic PRMs. For every fixed* k *and bounded Lipschitz path functionals* $\Phi_j:\mathcal{D}([0,T],S)\to\mathbb{R}$*,*

$$\mathbb{E}\left|\mathbb{E}\left[\prod_{j=1}^k\Phi_j(Y_j^N)\,\middle|\,\mathcal{F}_T^0\right]-\prod_{j=1}^k\int\Phi_j(y)\,\mathcal{P}^0(dy)\right|\longrightarrow 0. \tag{9}$$

Thus the factorization is conditional and holds in L^1*; unconditionally the limiting tagged paths generally remain dependent through the common artifact history.*

The endogeneity in Theorem 3.3 should not be obscured or exaggerated. The PRM is a convenient exogenous source of proposal points. Acceptance is population-generated because the predictable threshold is the current random quantity $\lambda_M a(\mu_{t-})$. The coupling uses the identity

$$\int_0^\infty\left|\mathbf{1}_{\{u\le x\}}-\mathbf{1}_{\{u\le y\}}\right|du=|x-y|.$$

This state-dependent thinning step is elementary. It is not presented as a new point-process construction.

The hybrid limit adds the surviving weak compensator:

$$\begin{aligned}\dot\mu_t&=F(\mu_t,B_t),\\ dB_t&=\{-\kappa B_t+\eta\lambda_w a(\mu_t)\}\,dt+h\,dJ_t,\\ J_t&=\int\mathbf{1}_{\{u\le\lambda_M a(\mu_{t-})\}}\Pi^0(dt,du).\end{aligned} \tag{10}$$

Theorem 3.4 (Hybrid information limit)**.** *Under the assumptions of Theorem 3.3, hybrid mode satisfies the same bound* (8) *with the limit* (10)*. The conditional factorization* (9) *also holds, with* $\mathcal{P}^0$ *defined using the hybrid board. Between common jumps the weak channel contributes deterministic board drift; at accepted artifact times the board has order-one common jumps.*

All three quantitative estimates follow from the shared stability and thinning argument in Appendix A, which explicitly controls the random finite-population routine board in hybrid mode. They concern the frozen production technology. The controlled-feedback statement in Theorem 7.2 is formulated separately and does not acquire a quantitative hybrid rate by citation to this theorem.

4 Access, readiness, and an exact benchmark

The probability limits leave production exogenous. Before introducing the full game, a two-state benchmark isolates how incentives select an artifact intensity. It uses the same access-and-readiness logic as the population game: uniform nontransferable opportunities give individual productivity proportional to $1/N$, while readiness is paid before assignment. It replaces the six-state task process and board by artifact absence or presence; it is a separate economic benchmark, not a state-space reduction of the full model.

4.1 Public information and admissible behavior

Let $Z \in \{0,1\}$ indicate useful-artifact availability. When $Z = 0$, player i chooses $u_i \in [0, \bar{u}]$ and the transition $0 \to 1$ has intensity $a_N \sum_i u_i$. When $Z = 1$, effort is required to be zero and $1 \to 0$ occurs at rate d. Initially $Z_0 = z$ is observed. Players observe the public state history, time, and their own controls. Current efforts are not an argument of rivals' strategies. A deviation preserves their prescribed rules. There is no unknown productivity or Bayesian posterior.

Throughout this section $\beta, \gamma > 0$, $d, v, \ell, a_N \geq 0$, and $\bar{u} < \infty$. The cost is $c(u) = \ell u + \gamma u^2/2$ and the payoff is

$$J_i(z) = \mathbb{E}_z \int_0^\infty e^{-\beta t}[vZ_t - c(u_i(t))]\,\mathrm{d}t.$$

Admissible deviations are predictable controls in the stated interval, using no future randomness. Intensities are bounded by $Na_N\bar{u} + d$ and discounted costs by $c(\bar{u})/\beta$, so all such controls are nonexplosive and integrable. The equilibrium concept is stationary Markov perfection; the verification below allows arbitrary admissible unilateral deviations. Uniqueness is claimed only within symmetric stationary Markov equilibria. A zero cap is the trivial game.

Write $k = \beta + d$ and $\mathfrak{a}_N = Na_N$. The normalized technology is $a_N = \lambda/N$; the fixed-productivity counterfactual is $a_N = \lambda$. These are different sequences of economic technologies, not two units for the same population experiment.

4.2 Solution with linear and quadratic costs and a cap

For a symmetric profile u, let $D = V_1 - V_0$. Its consistency equations are

$$\beta V_0 = -c(u) + \mathfrak{a}_N u D, \qquad \beta V_1 = v - dD, \qquad (k + \mathfrak{a}_N u)D = v + c(u). \tag{11}$$

A player's maximization treats the other $(N-1)$ efforts as fixed:

$$u = \mathrm{clip}_{[0,\bar{u}]}((a_N D - \ell)/\gamma). \tag{12}$$

This is a global maximizer because the Hamiltonian is strictly concave.

Proposition 4.1 (Complete symmetric solution)**.** *Assume $a_N > 0$ and $\bar{u} > 0$. If $a_N v \leq k\ell$, the unique symmetric stationary equilibrium has $u_N = 0$ and $D_N = v/k$. Otherwise let $\widehat{u}_N > 0$ be the unique positive solution of*

$$\gamma(N - \tfrac{1}{2})u^2 + \left(\frac{k\gamma}{a_N} + (N-1)\ell\right)u = v - \frac{k\ell}{a_N}. \tag{13}$$

Then $u_N = \min(\bar{u}, \widehat{u}_N)$ and

$$D_N = \frac{v + c(u_N)}{k + \mathfrak{a}_N u_N}, \qquad V_{N,1} = \frac{v - dD_N}{\beta}, \qquad V_{N,0} = \frac{v - kD_N}{\beta}.$$

The all-zero profile is an equilibrium exactly when $a_N v \leq k\ell$.

The average-welfare planner uses the same effort for all players in state zero. It is inactive exactly when $\mathfrak{a}_N v \leq k\ell$. Otherwise its unconstrained effort is the positive root of

$$\frac{\gamma}{2}u^2 + \frac{k\gamma}{\mathfrak{a}_N}u = v - \frac{k\ell}{\mathfrak{a}_N}, \tag{14}$$

and $u_{\mathrm{SP}} = \min(\bar{u}, \widehat{u}_{\mathrm{SP}})$. Its values follow (11) with that effort.

Proof. An interior Nash choice gives $D = (\ell + \gamma u)/a_N$; substituting into (11) gives (13). Its left side is strictly increasing for $u \geq 0$. At the cap the Kuhn–Tucker condition is

$$a_N \frac{v + c(\bar{u})}{k + \mathfrak{a}_N \bar{u}} \geq \ell + \gamma \bar{u}.$$

After multiplication by the positive denominator this is equivalent to $\widehat{u}_N \geq \bar{u}$. At zero the reverse marginal inequality gives the stated threshold. This exhausts the three cases and proves symmetric uniqueness. For a fixed sum of efforts, Jensen's inequality makes equal efforts cost minimizing under average welfare; hence the planner maximizes $\mathfrak{a}_N u D - c(u)$. Replacing a_N by $\mathfrak{a}_N$ in its marginal condition gives (14) and the cap rule.

To verify, for any unilateral predictable control x_t, apply the finite-state jump compensation formula to $e^{-\beta t} V_{Z_t}$ up to time H. The Bellman maximum implies

$$V_z \geq \mathbb{E}_z \left[\int_0^H e^{-\beta t} (v Z_t - c(x_t)) \, \mathrm{d}t + e^{-\beta H} V_{Z_H} \right].$$

Equality holds at the candidate effort. Boundedness of V eliminates the last term as $H \to \infty$. The identical calculation, after Jensen's inequality, verifies the planner against all joint predictable controls. This proves more than a first-order condition. □

When $a_N = 0$ or $v = 0$, zero effort is optimal; when the cap is zero the inactivity condition is vacuous. The positive-root function used in the code is $2s/(b + \sqrt{b^2 + 4as})$ for $au^2 + bu = s > 0$, avoiding cancellation.

4.3 Normalized quadratic benchmark and asymptotic orders

Set $\ell = 0$ and $a_N = \lambda/N$, with $\lambda, v > 0$. In the interior,

$$k D_N + \frac{\lambda^2}{\gamma} \left(\frac{1}{N} - \frac{1}{2N^2} \right) D_N^2 = v, \qquad u_N = \frac{\lambda D_N}{\gamma N}, \tag{15}$$

$$k D_{\mathrm{SP}} + \frac{\lambda^2}{2\gamma} D_{\mathrm{SP}}^2 = v, \qquad u_{\mathrm{SP}} = \frac{\lambda D_{\mathrm{SP}}}{\gamma}. \tag{16}$$

Both are positive roots. The Nash effort is at most the planner effort: the Nash polynomial in u has larger positive coefficients than the planner polynomial and the same constant surplus. At $N = 1$ the solutions coincide. From the planner equation $\gamma u_{\mathrm{SP}}^2/2 < v$, so $\bar{u} \geq \sqrt{2v/\gamma}$ suffices for both solutions to be interior. For smaller caps Proposition 4.1 applies; every fixed positive cap eventually ceases to bind at Nash equilibrium.

Put $L = \lambda^2 v/(\gamma k)$. With fixed positive primitives and $d > 0$,

Quantity	Symmetric Nash	Coordinated optimum
Individual effort (absent)	$\lambda v/(\gamma k N) + O(N^{-2})$	positive constant
Total effort (absent)	$\lambda v/(\gamma k) + O(N^{-1})$	$N u_{\mathrm{SP}}$
Discovery hazard (absent)	$L/N + O(N^{-2})$	$\lambda u_{\mathrm{SP}} > 0$
Stationary event frequency	$L/N + O(N^{-2})$	$d\lambda u_{\mathrm{SP}}/(d + \lambda u_{\mathrm{SP}})$
Stationary occupancy	$L/(dN) + O(N^{-2})$	$\lambda u_{\mathrm{SP}}/(d + \lambda u_{\mathrm{SP}})$
Value from state 0	$\lambda^2 v^2/(\beta \gamma k^2 N) + O(N^{-2})$	$W_0 > 0$

Here the planner column allows a binding cap. For an interior planner $W_0 = \lambda^2 D_{\mathrm{SP}}^2/(2\gamma\beta)$. The stationary net flow reward at constant effort and hazard q is

$$\overline{r} = \frac{vq - dc(u)}{d+q}.$$

This is distinct from the discounted value at a specified initial state. Under normalized Nash it is $vL/(dN) + O(N^{-2})$; under an active planner it is positive and independent of N (the planner marginal equation gives $vq > dc(u)$, also in the binding-cap case).

4.4 Waiting times, occupancy, and welfare baselines

For any of the stationary policies, put $q = \mathfrak{a}_N u$. Starting absent, the first discovery time is exponential with mean $1/q$ and $\mathbb{P}_0(\tau \le T) = 1 - e^{-qT}$, with mean infinity and probability zero if $q = 0$. Starting present, the time to a *new* discovery is the sum of independent exponentials of rates d and q. For $d, q > 0$ its mean is $1/d + 1/q$, and its survival function is

$$\mathbb{P}_1(\tau > T) = \begin{cases} (qe^{-dT} - de^{-qT})/(q-d), & q \neq d, \\ e^{-qT}(1+qT), & q = d. \end{cases} \tag{17}$$

If either rate is zero there is no new discovery from state one. If an initial artifact counts as "first useful information," that waiting time is instead zero. The occupancy probability solves $p' = q - (q+d)p$; for $q + d > 0$, $p(t) = q/(q+d) + (z - q/(q+d))e^{-(q+d)t}$.

The planner dominates the Nash profile from either initial state. Under normalized quadratic effort the limiting additive gaps are

$$W_0 - V_{N,0} \longrightarrow \frac{v - kD_{\mathrm{SP}}}{\beta} = W_0 > 0, \qquad W_1 - V_{N,1} \longrightarrow \frac{d}{\beta}\left(\frac{v}{k} - D_{\mathrm{SP}}\right). \tag{18}$$

Adding a constant flow reward b shifts every value by b/β and leaves these differences unchanged, while altering any welfare ratio. The initial-state-one gap vanishes at $d = 0$: a permanent initial artifact already provides the full reward.

For each finite N with positive Nash effort, discovery from absence eventually occurs almost surely. At $d = 0$, any positive hazard gives long-run occupancy one, although fixed-horizon discovery still vanishes under normalized Nash. At $d = q = 0$ every mixture of the two absorbing states is stationary. None of these stationary claims permits interchanging $T \to \infty$ and $N \to \infty$. Infinite discounted payoffs require $\beta > 0$; the finite-horizon population results below also permit $\beta = 0$. Sending both β and d to zero with N is a different asymptotic problem. With fixed $\beta > 0$, letting $\lambda \downarrow 0$ or $v \downarrow 0$ continuously removes effort and the gap.

4.5 Positive opportunity cost and the counterfactual

For normalized discovery, the all-zero Nash condition is $\lambda v/(Nk) \le \ell$, while planner activity requires $\ell < \lambda v/k$. Thus for any fixed $\ell \in (0, \lambda v/k)$ the planner is active but equilibrium is exactly inactive for every integer $N \ge \lambda v/(k\ell)$. Equation (13) supplies the positive-effort cases including a binding cap. The exact threshold is not merely an asymptotic order estimate.

For fixed individual productivity $a_N = \lambda$, quadratic costs instead give

$$kD_N + \frac{\lambda^2}{\gamma}(N - \tfrac{1}{2})D_N^2 = v, \qquad u_N = \lambda D_N/\gamma \sim \sqrt{v/(\gamma N)}, \qquad q_N \sim \lambda\sqrt{vN/\gamma}.$$

Hence aggregate discovery grows, occupancy tends to one when $d > 0$, stationary event frequency tends to d, and both Nash and planner values from zero tend to v/β. The planner's individual effort tends to $\min(\bar{u}, \sqrt{2v/\gamma})$ and its hazard is order N. For fixed $\ell > 0$, Nash is inactive for all N if $\lambda v \leq k\ell$; otherwise

$$u_N \sim \frac{v - k\ell/\lambda}{\ell N}, \qquad q_N \longrightarrow \lambda v/\ell - k > 0. \tag{19}$$

The planner becomes active once $N\lambda v > k\ell$ and its hazard again grows linearly for a positive cap. There is no aggregate collapse in either active fixed-productivity case. The linear-cost Nash limiting values are $V_0 = (v - k\ell/\lambda)/\beta$ and $V_1 = (v - d\ell/\lambda)/\beta$.

4.6 Fixed capacity and the allocation of access

The opportunity construction in Section 5 separates capacity from allocation. If each rate-$\lambda\bar{u}$ opportunity is independently assigned to i with exogenous probability π_i, and success probability is $u_i/\bar{u}$, the absent-state hazard is $\lambda \sum_i \pi_i u_i$. Uniform allocation gives $\pi_i = 1/N$ and the normalized technology. Keeping this same total opportunity stream but setting $\pi_1 = 1$ instead gives $q = \lambda u_1$; players without access choose zero costly readiness.

Proposition 4.2 (Designated access in the two-state benchmark)**.** *Fix $\lambda, v, \beta, \gamma, \bar{u} > 0$, $d, \ell \geq 0$. With all opportunities assigned to player 1, its unique stationary optimal effort is zero exactly when $\lambda v \leq (\beta + d)\ell$. Otherwise it is*

$$u_D = \min(\bar{u}, \widehat{u}_D) > 0, \qquad \frac{\gamma}{2}\widehat{u}_D^2 + \frac{k\gamma}{\lambda}\widehat{u}_D = v - \frac{k\ell}{\lambda}, \qquad k = \beta + d.$$

Its private incentive and discovery rate $q_D = \lambda u_D > 0$ are independent of N.

Proof. The creator's Bellman equations give $(k + \lambda u_D)D = v + c(u_D)$ and $u_D = \mathrm{clip}_{[0,\bar{u}]}((\lambda D - \ell)/\gamma)$. These are exactly the one-player equations of Proposition 4.1, including its cap rule and verification. Other players have no effect on discovery and strictly prefer zero effort. □

Thus fixed capacity alone does not imply collapse. Concentrating access changes the institution and does not refute the uniform-allocation theorem. It neither derives endogenous hierarchy nor proves delegation efficient, stable, implementable, or first-best in the full model. In particular, the creator still values only its own reward; average welfare counts its cost as $c(u_D)/N$. Convex costs favor spreading a fixed total effort only when marginal productivities are equal. Under the original uniform allocation, restricting production to K_N agents without reallocating their access still gives $\Lambda_N \leq \lambda_M \bar{u} K_N/N$.

4.7 A benchmark scaling classification

Proposition 4.3 (Conditional discovery hazard under productivity scaling)**.** *In the symmetric stationary two-state game let $a_N = \lambda N^{-s}$, $s \geq 0$, with fixed $\lambda, v, \beta, \gamma, \bar{u} > 0$, $d \geq 0$, and $\ell = 0$. Every fixed positive cap is eventually inactive. Let $D_c > 0$ solve $kD_c + \lambda^2 D_c^2/\gamma = v$. Then:*

Regime	*Individual effort u_N*	*Absent-state hazard $q_N = N a_N u_N$*
$0 \leq s < 1/2$	$\sim \sqrt{v/\gamma}\, N^{-1/2}$	$\sim \lambda\sqrt{v/\gamma}\, N^{1/2-s} \to \infty$
$s = 1/2$	$\sim (\lambda D_c/\gamma) N^{-1/2}$	$\to \lambda^2 D_c/\gamma > 0$
$s > 1/2$	$\sim (\lambda v/(\gamma k)) N^{-s}$	$\sim (\lambda^2 v/(\gamma k)) N^{1-2s} \to 0$

For fixed $\ell > 0$, every $s > 0$ instead gives eventual exact inactivity. At $s = 0$ the active region is $\lambda v > k\ell$ and $q_N \to \lambda v/\ell - k > 0$, as in (19).

Proof. Without the cap the positive value-gap root satisfies

$$kD_N + \frac{\lambda^2}{\gamma}(N^{1-2s} - \tfrac{1}{2}N^{-2s})D_N^2 = v, \qquad u_N = \lambda N^{-s} D_N/\gamma.$$

For $s > 1/2$, $D_N \to v/k$; at $s = 1/2$, $D_N \to D_c$; for $s < 1/2$, $D_N \sim \sqrt{\gamma v}\, N^{s-1/2}/\lambda$ by dividing by the dominant quadratic coefficient. All three expressions give $u_N \to 0$, proving eventual cap inactivity and the table. With linear cost, the exact inactivity condition $\lambda N^{-s} v \le k\ell$ eventually holds for every $s > 0$. □

For $d > 0$, occupancy is $q_N/(d+q_N)$ and stationary event frequency is $dq_N/(d+q_N)$. A diverging absent-state hazard therefore gives occupancy tending to one and event frequency tending to d, not infinity. For $d = 0$, positive discovery leads to absorbing occupancy one and long-run event frequency zero. Zero productivity, benefit, or cap removes activity; discount is fixed strictly positive, so joint vanishing-discount limits are excluded. The chosen $s = 1$ lies within a broader incentive-collapse region. This is a benchmark classification, not a six-state theorem or a claim of a new phase transition. Large population size alone selects none of these regimes.

5 The production game and finite-N equilibrium

5.1 Controls, information, and the access institution

Retain the state space, rates $q_{yy'}(b)$, routine channel, and board decay from Section 2. Only artifact production is now chosen:

$$\Lambda^N(t, \boldsymbol{y}, b) = \frac{\lambda_M}{N} \sum_{i=1}^{N} \mathbf{1}_{\{c_i = \mathsf{A}\}} u_i(t, \boldsymbol{y}, b), \qquad 0 \le u_i \le \bar{u}. \tag{20}$$

An artifact still adds h without changing a private state. Effort has no other dynamical or direct private benefit. Exogenous unit production in Section 3 is neither asserted to be an equilibrium nor admissible here when $\bar{u} < 1$. A coordinated policy below must satisfy the game's cap and eligibility constraints. Creator marks become relevant only for the transfers in Section 8.

One explicit institutional interpretation is a fixed stream of experimental opportunities at rate $\lambda_M \bar{u}$, each assigned uniformly among the N players. Effort represents costly readiness chosen before the assignment: an assigned eligible player succeeds with probability $u_i/\bar{u}$, and an ineligible assignment expires unused. Independent thinning gives exactly (20). Readiness costs accrue continuously, whether or not an opportunity arrives. The institution is a fixed opportunity budget with exogenously dispersed access, non-reassignable allocations, and readiness costs incurred before assignment. Fixed capacity bounds total proposals; uniform access makes an individual's marginal success rate λ_M/N; expiration prevents redirecting unused assignments to an eligible prepared player. Paying for effort only after assignment would change the cost calculation. The unit of readiness, its cap, and (ℓ, γ) stay fixed across N, as do per-person rewards; welfare is their average. This is not a change of units or a fixed total benefit divided among more people.

A facility issuing nontransferable lottery slots to users who must maintain readiness is an illustrative interpretation. Reassignable access, a designated operator, or preparation after allocation may make it a poor description. No evidence supplied here establishes these restrictions in the

motivating AI-agent setting. Section 4.6 isolates allocation within the two-state benchmark. The zero-cap case has no opportunities and needs no division by $\bar{u}$.

For the full game all labels' states and the board are publicly observed. The state for arbitrary asymmetric feedbacks is $s = (\boldsymbol{y}, b) \in S^N \times \mathbb{R}_+$. The generator is

$$\begin{aligned}\mathcal{L}_N^u f(s) = & -\kappa b \partial_b f(s) + \sum_i \sum_{y'} q_{y_i y'}(b)[f(s^{i\to y'}) - f(s)] \\ & + w\lambda_w \sum_i \mathbf{1}_{\{c_i=\mathsf{A}\}}[f(\boldsymbol{y}, b+\eta/N) - f(s)] + \Lambda^N(s)[f(s^h) - f(s)]. \end{aligned} \tag{21}$$

For symmetric feedback $u_i = \theta(t, y_i, \mu, b)$, the empirical generator is (1) with its artifact term replaced by

$$\lambda_M \sum_{y:C(y)=\mathsf{A}} \mu(y)\theta(t, y, \mu, b)[f(\mu, b+h) - f(\mu, b)].$$

Thus the common architecture is unchanged; the accepted intensity changes. A tagged player's value needs (y_i, μ, b) even under symmetry. Its generator separates its own rate-q jump (changing both y_i and μ) from rival jumps with counts $N\mu(y) - \mathbf{1}_{\{y_i=y\}}$. An empirical law alone loses the tagged state; with asymmetric rules it can also lose the population's Markov property.

Fix $T < \infty$, $\beta \geq 0$, $\gamma > 0$, $\ell \geq 0$, $0 \leq \bar{u} < \infty$, and zero terminal values. For remaining-horizon discounting,

$$J_i(t, s; u) = \mathbb{E}_{t,s} \int_t^T e^{-\beta(r-t)}[g(Y_i(r)) - \mathbf{1}_{\{C_i(r)=\mathsf{A}\}} c(u_i(r))]\, \mathrm{d}r, \qquad 0 \leq g \leq R.$$

The substantive welfare result uses $g(x, c) = Rx$ and $c_1 > c_0$. Controls are predictable for the usual augmentation of the primitive-clock filtration, optionally enlarged by independent randomization. Future clock increments and proposal marks are unavailable when a control is chosen; ineligible effort is zero. The bound (3) with $L_0 = \lambda_M \bar{u}$ proves nonexplosion for these controls by the same interlacing construction. Costs are bounded, so integrability is automatic. A specified nonanticipative control rule determines an adapted process; time-dependent Markov feedback has the strong Markov restart property after time is included in the state. The equilibria below are verified against predictable unilateral deviations. Rivals keep their state-feedback rules; their controls do not respond instantaneously to an observed deviating current effort.

5.2 Direct continuous-board existence and verification

The continuous board cannot be ignored in an equilibrium-existence argument. Here its unbounded exposure rate belongs entirely to an *uncontrolled* baseline process. The controlled perturbation has bounded rate and one board-jump operator. This special structure closes the finite-N problem.

Let P_r^0 denote the transition semigroup on $S^N \times \mathbb{R}_+$ with the rare channel disabled, retaining routine posts when $w = 1$. It is a positive contraction on bounded functions. It maps bounded continuous functions to bounded continuous functions and is jointly pointwise continuous in (r, s). To check this, restrict initial boards to a compact set and condition on routine proposal count. On each finite-count event the board stays bounded and individual rates are locally Lipschitz. A shared-clock coupling proves continuity; the Poisson tail permits dominated passage to unrestricted counts. This also gives stochastic continuity and compact containment locally uniformly in the initial board. On a bounded initial-board set and a bounded routine proposal count, the paths stay in a compact set. Continuous test functions are uniformly continuous there; the omitted Poisson tail costs at most twice their bound times its probability. This applies also to bounded jointly

continuous tests $f(t,s)$ with varying time argument. Independent increments and the pathwise construction give the strong Markov restart property used in Appendix B.1. Global sup-norm strong continuity of P^0 is neither assumed nor needed.

For a vector of bounded continuous functions $V = (V_1, \dots, V_N)$ define

$$\begin{aligned} z_i(s) &= V_i(s^h) - V_i(s), \qquad A_i[V](s) = \mathbf{1}_{\{c_i = \mathsf{A}\}} \operatorname{clip}_{[0,\bar{u}]}\left(\frac{(\lambda_M/N) z_i(s) - \ell}{\gamma}\right), \\ \Lambda[V](s) &= \frac{\lambda_M}{N} \sum_i A_i[V](s), \qquad L = \lambda_M \bar{u}, \\ G_i[V](s) &= g(y_i) - c(A_i[V](s)) + (L - \Lambda[V](s)) V_i(s) + \Lambda[V](s) V_i(s^h). \end{aligned} \tag{22}$$

Since $c(0) = 0$, this includes ineligible players. On $[a,b]$ with terminal vector ψ, the precise HJB notion used here is the killed-semigroup equation

$$V_i(t) = e^{-(\beta+L)(b-t)} P^0_{b-t} \psi_i + \int_t^b e^{-(\beta+L)(r-t)} P^0_{r-t} G_i[V(r)] \, \mathrm{d}r. \tag{23}$$

It requires no board derivatives. Equivalently, its un-killed version has discount β and driver $g_i - c(A_i[V]) + \Lambda[V] \Delta_h V_i$.

Theorem 5.1 (Finite-N continuous Markov equilibrium)**.** *For every finite N, T and the above primitives, (23) with terminal zero has a unique bounded jointly continuous solution on $[0,T] \times S^N \times \mathbb{R}_+$. Its feedback $A[V]$ is a pure continuous Markov-perfect equilibrium, verified against bounded predictable unilateral deviations. Values are unique within this bounded continuous mild-HJB class; controls are its unique Hamiltonian maximizers. Permutation invariance makes the constructed equilibrium symmetric. No compact-board truncation, classical derivatives, relaxed controls, or small-total-horizon restriction is imposed. Uniqueness outside this verification class or among history-based equilibria is not asserted.*

Proof. Use the maximum over indices and states as the vector norm. On $\|V\|, \|W\| \le M$, with $d_* = \|V - W\|$,

$$\|A_i[V] - A_i[W]\| \le \frac{2\lambda_M}{\gamma N} d_*, \qquad \|\Lambda[V] - \Lambda[W]\| \le \frac{2\lambda_M^2}{\gamma N} d_*.$$

Cost is $(\ell + \gamma\bar{u})$-Lipschitz on the effort interval. The weights $L - \Lambda$ and Λ are nonnegative and sum to L. Thus a Lipschitz constant for G on this ball is

$$C_M = L + \frac{4M\lambda_M^2 + 2\lambda_M(\ell + \gamma\bar{u})}{\gamma N}, \tag{24}$$

and $\|G[V]\| \le R + c(\bar{u}) + LM$. Both can be bounded independently of N. Let $K_t = R \int_0^t e^{-\beta r} \, \mathrm{d}r$, set $M = K_T + 1$, and take continuous terminal values in $[0, K_T]$. Work in the closed radius-M ball of $C_b([a,b] \times S^N \times \mathbb{R}_+)^N$ with terminal slice ψ, using the supremum norm; this is complete. Put $B_M = R + c(\bar{u}) + LM$ and let $\widehat{C}_M$ be (24) with $1/N$ replaced by 1. A slab length

$$b - a \le \min\{[2(B_M + 1)]^{-1}, [2(\widehat{C}_M + 1)]^{-1}\}$$

makes the operator norm at most $K_T + 1/2 < M$ and its contraction factor less than $1/2$. It preserves the terminal slice. Compact containment and bounded dominated convergence give joint

continuity, including at the endpoints. Banach iteration gives a local solution with a slab length independent of N; no bound uniform over unbounded T is asserted.

Verification supplies the bound needed to continue. For a local solution use the un-killed driver $F_i = g_i - c(A_i) + \Lambda[A]\Delta_h V_i$. The mild identity and the Markov property give the baseline martingale

$$\begin{aligned} &e^{-\beta(r-t)}V_i(r, S^0_r) + \int_t^r e^{-\beta(s-t)}F_i(s, S^0_s)\,\mathrm{d}s \\ &\quad = \mathbb{E}\left[e^{-\beta(b-t)}\psi_i(S^0_b) + \int_t^b e^{-\beta(s-t)}F_i(s, S^0_s)\,\mathrm{d}s \mid \mathcal{F}_r\right]. \end{aligned}$$

Appendix B.1 proves killed/un-killed equivalence and the interlacing identity at random proposal times with predictable acceptance. In particular, under an actual unilateral deviation,

$$e^{-\beta(r-t)}V_i(r, S_r) + \int_t^r e^{-\beta(s-t)}[F_i(s, S_{s-}) - \Lambda^{\text{actual}}_s \Delta_h V_i(s, S_{s-})]\,\mathrm{d}s$$

is a true martingale. This uses baseline restarts and compensation, without assuming differentiability of V_i.

Relative to the candidate, a deviator's instantaneous residual is

$$\frac{\lambda_M}{N}(u_i - A_i)\Delta_h V_i - c(u_i) + c(A_i) \le 0,$$

since A_i is the global maximizer of the strictly concave Hamiltonian. Consequently V_i dominates every deviating payoff with terminal ψ_i, with equality at the candidate. Starting with zero terminal payoff and pasting backward verifies every constructed interval. At the next slab's terminal time b, the already verified continuation obeys $0 \le \psi_i \le K_{T-b}$. Zero effort guarantees nonnegative reward and continuation under fixed rival feedback. Nonnegative costs give the upper bound $K_{b-t} + e^{-\beta(b-t)}K_{T-b} = K_{T-t}$:

$$0 \le V_i(t, s) \le K_{T-t} \le K_T. \tag{25}$$

Hence the same ball and interval length can be used repeatedly. Finitely many steps cover $[0, T]$. The pasted solution satisfies the global mild identity by the semigroup property. Two bounded solutions lie in some common finite ball; contraction uniqueness on successive short slabs makes them identical. Finally, label permutations preserve all equations, proving symmetry. □

This is a class-specific existence result. It works because the unique local maximizer depends only on that player's jump-value difference: no unresolved simultaneous action selection is hidden in the coupled driver.

5.3 Uniform incentives, including asymmetric verified equilibria

Corollary 5.2 (Small private incentives)**.** *Every Markov equilibrium with bounded continuation values satisfying the local verification identity obeys*

$$u_i^{\text{NE}} = \text{clip}_{[0,\bar{u}]}\left(\frac{(\lambda_M/N)\Delta_h V_i - \ell}{\gamma}\right) \le \min\left(\bar{u}, \frac{(\lambda_M K_T/N - \ell)_+}{\gamma}\right) \tag{26}$$

at eligible states. In particular, for $\ell = 0$,

$$u_i^{\text{NE}} \le \frac{\lambda_M K_T}{\gamma N}, \qquad \Lambda_N^{\text{NE}} \le \frac{\lambda_M^2 K_T}{\gamma N} =: \frac{C_*}{N}.$$

For $\ell > 0$, all equilibrium exploration is zero whenever $\lambda_M K_T/N \le \ell$. This is sufficient, not a full characterization of activity below that threshold. Symmetry is unnecessary for the estimate.

Proof. The zero-control deviation gives the lower bound in (25); nonnegative costs give the upper bound. Thus $\Delta_h V_i \le K_T$. Insert this in the Hamiltonian maximizer and sum over at most N eligible players. The tighter bound K_{T-t} is available at time t. □

An artifact may change many future transitions and rivals' future feedback responses. All those effects are inside $V_i(s^h) - V_i(s)$, whose positive part cannot exceed the player's bounded total reward. No uniform board derivative or weak amplification hypothesis is required. Unbounded transfers and direct private effects of effort invalidate the premises.

5.4 The seed and the finite planner

If $B_0 = 0$ and all agents are unexposed, with no exposed or adopter seed, the communication system remains unexposed under every policy, also in hybrid mode. There is nobody eligible to produce the board needed for exposure. Task completion at c_0 can continue; it is the communication configuration that is absorbing.

For reference, the finite planner's formal empirical-state equation is

$$0 = \partial_t W_N + \mathcal{L}_N^0 W_N - \beta W_N + \langle \mu, g \rangle + a(\mu) \max_{0 \le u \le \bar{u}} \{\lambda_M u \Delta_h W_N - c(u)\}. \tag{27}$$

For a fixed total eligible effort, equal eligible effort minimizes cost and leaves the state law unchanged. Thus this scalar maximization is valid despite different private reward states. Its maximizer is $\mathrm{clip}_{[0,\bar{u}]}((\lambda_M \Delta_h W_N - \ell)/\gamma)$. The same bounded-perturbation argument gives a unique continuous bounded mild solution and a verified finite-N optimal feedback: the scalar maximized Hamiltonian is Lipschitz in its jump difference with constant $\lambda_M \bar{u}$; verification gives $0 \le W_N \le K_T$ and continuation. This establishes finite-horizon existence, not an explicit optimal solution or convergence of optimal policies. The following welfare proof needs only the planner supremum.

6 A uniform, strictly positive welfare improvement

6.1 Performance difference with an explicit remainder

Let $W_N^0(t, \mu, b)$ be average reward under zero exploration, with the same routine channel as the comparison system. It is a bounded baseline potential. Let eligible agents exert $\varepsilon\chi(t, \mu, b)$, where $0 \le \chi \le 1$ is Borel and predictable when evaluated at left limits, and $0 \le \varepsilon \le \bar{u}$. Let W_N^ε be the resulting welfare from the given initialization, including exploration costs. Write

$$A_N = \mathbb{E}^0 \int_0^T e^{-\beta t} a(\mu_t^N) \chi(t, \mu_t^N, B_t^N) [\lambda_M \Delta_h W_N^0(t, \mu_t^N, B_t^N) - \ell] \, \mathrm{d}t. \tag{28}$$

Theorem 6.1 (Uniform policy variation)**.** *For every N and any such gate,*

$$W_N^\varepsilon - W_N^0 = \varepsilon A_N + \mathcal{R}_N(\varepsilon), \qquad |\mathcal{R}_N(\varepsilon)| \le \left[2\lambda_M(\lambda_M K_T + \ell) T^2 + \frac{\gamma T}{2} \right] \varepsilon^2. \tag{29}$$

The same bound applies in rare-only and hybrid mode. No monotonicity or regularity of the value as a function of the board is needed for this identity.

Proof. The baseline potential identity, interlaced at bounded-rate artifact proposals as in Theorem 5.1, gives the exact performance difference

$$W_N^\varepsilon - W_N^0 = \mathbb{E}^\varepsilon \int_0^T e^{-\beta t} \left\{ \varepsilon a \chi (\lambda_M \Delta_h W_N^0 - \ell) - \frac{\gamma \varepsilon^2}{2} a \chi^2 \right\} \mathrm{d}t. \tag{30}$$

Terminal values are zero. This accounts for the current eligible fraction, the λ_M discovery productivity, subsequent reward gain, discounting, and both cost components.

Couple the controlled and baseline systems with identical initial data and private/routine clocks. Add an independent artifact proposal process of rate $\lambda_M\varepsilon$, accepting with probability $a\chi$. Before the first accepted artifact all states coincide exactly. The probability of any accepted artifact is at most $\lambda_M\varepsilon T$. The integrand $Q = a\chi(\lambda_M\Delta_h W_N^0 - \ell)$ is bounded in absolute value by $M = \lambda_M K_T + \ell$. Therefore

$$\left|\mathbb{E}^\varepsilon \int_0^T e^{-\beta t} Q_t \,\mathrm{d}t - \mathbb{E}^0 \int_0^T e^{-\beta t} Q_t \,\mathrm{d}t\right| \le 2MT\,\lambda_M\varepsilon T.$$

The quadratic cost term is bounded by $\gamma\varepsilon^2 T/2$. Substitution proves (29). The number of accepted events is dominated by $P(\lambda_M\varepsilon T)$, and $\mathbb{P}(J_T \ge 2) \le (\lambda_M\varepsilon T)^2/2$ by its second factorial moment. Thus multiple artifacts are accounted for rather than suppressed in the first-order formula. □

6.2 A productive region with an explicit witness

Take rare-only mode, $B_0 = 0$, $\bar{u}, R, \rho, \alpha, h, \lambda_M > 0$, $c_1 > c_0$, and fixed finite nonnegative remaining rates. Initial deterministic counts satisfy

$$a(\mu_0^N) \ge a_* > 0, \qquad \mu_0^N(0, \mathsf{U}) \ge m_* > 0. \tag{31}$$

Counts may be randomly permuted independently of future clocks. Choose $0 < t_0 < T$ and $\chi = \mathbf{1}_{[0,t_0)}$. Endpoint choice has no effect on payoffs.

Lemma 6.2 (Rare-only zero-board monotonicity)**.** *With all future artifact production disabled, a forced board jump from zero at time t weakly increases expected task reward from every private initial state. For a player in $(0, \mathsf{U})$ at that time and $t \le t_0$, its remaining value increase is at least $L_* > 0$, uniformly in N and in the other players' states.*

Proof. Without a board, U is absorbing for the communication coordinate. Couple baseline and forced-artifact communication chains identically until their first entry into U. This entry is simultaneous because rates out of E, A do not depend on the board. Thereafter the baseline stays U. Before that entry the two task completion intensities agree; afterwards the forced-artifact chain's completion intensity is always either c_0 or c_1, while the baseline's is c_0.

Use a common rate-c_0 clock that sets incomplete tasks to complete, a common rate-r clock that resets completed tasks, and an independent rate-$(c_1 - c_0)$ completion clock accepted when the corresponding communication state is A. The task coordinate in the forced-artifact chain remains at least the baseline coordinate. Communication states themselves need not be globally ordered. This proves nonnegative reward differences for every initial state.

For strictness put $s_* = (T - t_0)/4$, $b_* = he^{-\kappa T}$, and

$$\begin{aligned} q_* &= e^{-(c_0+r+\nu+\delta)T}(1 - e^{-\rho b_* s_*})(1 - e^{-\alpha s_*})(1 - e^{-(c_1-c_0)s_*}), \\ L_* &= Rs_* e^{-\beta T} q_* > 0. \end{aligned} \tag{32}$$

Start both chains at $(0, \mathsf{U})$ immediately after the forced artifact time. Forbid all potential c_0 completion, reset, forgetting, and abandonment clocks on the following four slots of length s_*. This has probability at least $e^{-(c_0+r+\nu+\delta)T}$; these are independent primitive Poisson clocks, including their rings when their transition is disabled. Require at least one rate-ρb_* exposure proposal in slot one, one adoption-clock ring in slot two, and one extra-completion ring in slot three. Exposure is feasible because the decaying forced board remains at least b_* throughout these slots. If a required

favorable transition has already occurred, the forbidden clocks ensure it persists, so additional favorable rings do not invalidate the witness. The three independent positive probabilities multiply as in (32). The forced task is complete and the baseline task incomplete throughout slot four, yielding discounted reward at least $Rs_*e^{-\beta T}$. All other reward differences are nonnegative by the coupling. □

The witness uses independent potential clocks, including disabled rings, and the absorbing zero-board baseline. It supplies no global order of communication states at arbitrary positive boards.

Theorem 6.3 (Uniform coordination gain)**.** *Under the preceding productive-region assumptions, including $\bar{u} > 0$, (31), rare-only mode and $\ell = 0$,*

$$A_N \geq A_* := \lambda_M L_* a_* m_* \int_0^{t_0} e^{-(\beta+\delta+c_0)t}\,\mathrm{d}t > 0 \tag{33}$$

for every admissible population size. Write $C = 2\lambda_M^2 K_T T^2 + \gamma T/2$ and choose $0 < \varepsilon \leq \min(\bar{u}, A_*/(2C))$. Then

$$W_N^{\mathrm{SP}} \geq W_N^{\varepsilon} \geq W_N^0 + \delta_*, \qquad \delta_* = \varepsilon A_*/2 > 0.$$

For the equilibrium from Theorem 5.1, and for any other verified Markov equilibrium satisfying Corollary 5.2,

$$\liminf_{N\to\infty}(W_N^{\mathrm{SP}} - W_N^{\mathrm{NE}}) \geq \delta_*.$$

Proof. Along the zero-exploration law the board is zero and agents are independent given their deterministic initial labels. By Lemma 6.2, $\Delta_h W_N^0(t,\mu,0) \geq L_*\mu(0,\mathsf{U})$ for $t \leq t_0$. Let I be the set of initial adopters and J the disjoint set of initial $(0,\mathsf{U})$ players. An $i \in I$ that has no abandonment ring before t remains an adopter, with probability $e^{-\delta t}$. A $j \in J$ with no baseline completion ring remains $(0,\mathsf{U})$, with probability $e^{-c_0 t}$. Their clocks are independent. Consequently

$$\mathbb{E}^0[a(\mu_t^N)\mu_t^N(0,\mathsf{U})] \geq \frac{|I||J|}{N^2} e^{-(\delta+c_0)t} \geq a_* m_* e^{-(\delta+c_0)t}.$$

Insert this in (28); random permutation of the initial groups does not change the argument. The variation bound gives $W_N^{\varepsilon} - W_N^0 \geq \varepsilon A_* - C\varepsilon^2 \geq \delta_*$. The final inequality uses the equilibrium baseline convergence proved next. The planner supremum exceeds the feasible policy whether or not one uses its finite-N existence result. □

With linear costs, the same region gives $A_N^{(\ell)} \geq A_* - \ell t_0$, so a sufficiently small positive $\ell < A_*/t_0$ retains a strictly positive coordinated gain. The constant in (29) then includes ℓ. This is conservative but concrete, and large-N equilibrium is exactly inactive.

The proof does not assert positive planner effort at every state or close to the terminal time. It fails without a seed, without task benefit, or if the linear marginal cost is too large. For positive initial boards and for hybrid posting, (29) remains valid; the positivity theorem does not automatically carry over. A fresh positivity argument or the explicit condition $\liminf A_N > 0$ is needed there.

7 Disappearance versus a common-jump limit

7.1 Equilibrium: probability, strong coupling, and welfare

Set $\ell = 0$ initially and let $C_* = \lambda_M^2 K_T/\gamma$. The artifact count at any verified equilibrium satisfies

$$\mathbb{P}(J_T^{N,\mathrm{NE}} \geq 1) \leq \mathbb{E}J_T^{N,\mathrm{NE}} \leq C_* T/N. \tag{34}$$

Couple the equilibrium process to its rare-disabled version with identical initial state and identical individual and routine driving clocks. Effort changes no other transition, so all states agree exactly until the first rare artifact, even if rivals' future feedback reactions are highly state dependent. If Φ is any bounded functional of the whole population path (including any fixed collection of tagged paths), then

$$|\mathbb{E}\Phi(S^{N,\mathrm{NE}}) - \mathbb{E}\Phi(S^{N,0})| \le 2\|\Phi\|_\infty C_* T/N. \tag{35}$$

This is a bounded path-test estimate; it is not by itself a bound for the unbounded board norm.

Proposition 7.1 (Strong extra error and welfare convergence)**.** *Suppose $B_0^N \le K$ almost surely. There is a coupling such that, in both rare-only and hybrid modes,*

$$\mathbb{E}\sup_{t\le T}(\|\mu_t^{N,\mathrm{NE}} - \mu_t^{N,0}\|_1 + |B_t^{N,\mathrm{NE}} - B_t^{N,0}| + J_t^{N,\mathrm{NE}}) \le C_T/N.$$

Writing average equilibrium and no-exploration welfare as W_N^{NE}, W_N^0,

$$|W_N^{\mathrm{NE}} - W_N^0| \le K_T C_* T/N + \frac{\lambda_M^2 K_T^2 T}{2\gamma N^2}. \tag{36}$$

For $\ell > 0$, all these differences are exactly zero for sufficiently large N.

Proof. Construct a common routine proposal process of rate $N\lambda_w$ with uniform label marks and an independent rare proposal process P_N of rate C_*/N. The first is valid for both populations; accept routine proposals when the marked agent is eligible. The second dominates the equilibrium rare intensity. Let $E_N = \{P_N(T) \ge 1\}$. Off E_N, the coupled paths coincide. On E_N, empirical error is at most 2, and joint board domination gives

$$\sup|B^{N,\mathrm{NE}} - B^{N,0}| \le 2K + \frac{2\eta}{N}P_w(N\lambda_w T) + hP_N(T).$$

The routine proposal count is independent of E_N. Therefore multiplication by $\mathbf{1}_{E_N}$ and expectation bounds the right side by $[2K + 2\eta\lambda_w T]\mathbb{P}(E_N) + hC_*T/N$. This proves the board estimate without a Hölder loss; the empirical and count estimates are immediate. The argument works with initial random boards bounded by K, without requiring independence of the initial populations from each other in this comparison. All future proposal clocks are independent of initial data.

Each player's total discounted task reward lies in $[0, K_T]$; on the coupling it can differ only on E_N. Average quadratic effort cost is at most $\gamma T(\lambda_M K_T/(\gamma N))^2/2$. Combining the two yields (36). If $\ell > 0$, Corollary 5.2 disables the rare channel exactly for large N, and effort cost is also zero. □

Under the standing initialization (4), the artifact-disabled limit is precisely (5), with $w = 0$ or 1. Theorem 3.2 and Proposition 7.1 give expected uniform error $C_T(\varepsilon_{N,p} + N^{-1/2})$ from equilibrium to that deterministic system. This composes the probability foundation with the incentive estimate rather than assuming equilibrium production exogenous.

Under exchangeable chaotic initialization, the same coupling and the tagged argument in Lemma A.3 give ordinary tagged path independence. Hybrid equilibrium removes rare common jumps and leaves the deterministic routine background. Common random initialization or unrelated imposed common noise would remain; they are excluded by the deterministic initialization used in this statement.

7.2 Coordination under a fixed feedback

Fix, independently of N, eligible feedback $\theta(t, y, \mu, b) \in [0, \bar{u}]$, zero when $C(y) \neq \mathsf{A}$. Assume it is Borel in time and uniformly Lipschitz in (μ, b), with the same Lipschitz constant for all t, y. Finite deterministic time discontinuities, including the early gate, are allowed. Then

$$\mathcal{A}(t, \mu, b) = \lambda_M \sum_{y: C(y)=\mathsf{A}} \mu(y)\theta(t, y, \mu, b)$$

is bounded by $L = \lambda_M \bar{u}$ and globally Lipschitz in (μ, b). On a shared PRM $\Pi^0(\,\mathrm{d}t,\,\mathrm{d}z)$ of intensity $\mathrm{d}t\,\mathrm{d}z$, the candidate limit is

$$\begin{aligned} \dot{\mu}_t &= F(\mu_t, B_t), \\ \mathrm{d}B_t &= \{-\kappa B_t + w\eta\lambda_w a(\mu_t)\}\,\mathrm{d}t + h\,\mathrm{d}J_t, \\ J_t &= \int_{(0,t]\times[0,L]} \mathbf{1}_{\{z \le \mathcal{A}(s, \mu_{s-}, B_{s-})\}} \Pi^0(\,\mathrm{d}s,\,\mathrm{d}z). \end{aligned} \tag{37}$$

It has a unique pathwise construction: solve the locally Lipschitz deterministic flow between finitely many proposals and decide acceptance from the left limit. The empirical law is continuous even though the common board jumps.

Theorem 7.2 (Fixed-policy convergence)**.** *Under the preceding feedback assumptions and initialization* (4)*, in rare-only mode the finite system using the same feedback and common PRM satisfies*

$$\mathbb{E}\sup_{t \le T}(\|\mu_t^N - \mu_t\|_1 + |B_t^N - B_t| + |J_t^N - J_t|) \le C_T(\varepsilon_{N,p} + N^{-1/2}).$$

In hybrid mode the same coupled triple converges in expected uniform distance under these assumptions. The localization proof given here does not assert a quantitative hybrid rate for the general controlled feedback. With exchangeable chaotic initialization, for bounded Lipschitz path functionals $\Phi_1, \ldots, \Phi_k$*,*

$$\mathbb{E}\left|\mathbb{E}\left[\prod_{j=1}^{k} \Phi_j(Y_j^N) \mid \mathcal{F}_T^0\right] - \prod_{j=1}^{k} \int \Phi_j(y)\mathcal{P}^0(\,\mathrm{d}y)\right| \longrightarrow 0, \tag{38}$$

where $\mathcal{F}_T^0 = \sigma(\Pi^0|_{[0,T]\times[0,L]})$ *and* $\mathcal{P}^0$ *is the conditional law of a chain driven by independent private clocks in* B*.*

Proof. For rare-only mode, condition on the number m and times of common proposals, but not on their independent uniform height marks. Both boards are bounded by $K + hm$. The private martingale has conditional L^p supremum bounded by $C(1+m)^q/\sqrt{N}$, using its bracket bound $C(1 + K + hm)T/N$ and its $O(1/N)$ jump sizes. Conditioning on the proposal schedule leaves the initial error distribution unchanged. Private PRMs retain their intensities in the filtration enlarged by that schedule, and height marks are revealed only at their proposals. Stop just before the first acceptance mismatch. Up to this time, common jumps agree and the drift comparison yields a Gronwall factor at most $e^{C(1+m)T}$. At any proposal the conditional mismatch probability is exactly $|\mathcal{A}^N - \mathcal{A}|/L$, bounded by a fixed constant times the pre-proposal state error. A union bound over the m marks thus gives conditional first-mismatch probability at most $C(1+m)^q e^{cm}(\varepsilon_{N,p} + N^{-1/2})$. After a mismatch the population error is at most 2, and the board/count error at most a constant times $1 + m$. Poisson exponential moments sum this bound. The case $L = 0$ is the baseline LLN.

For hybrid convergence, stop both systems when either board first exceeds H. On the stopped domain all drift coefficients have a deterministic Lipschitz constant C_H and the compensated mismatch identity is

$$\mathbb{E}K^N_{t\wedge\tau_H} = \mathbb{E}\int_0^{t\wedge\tau_H} |\mathcal{A}(s,\mu^N_s,B^N_s) - \mathcal{A}(s,\mu_s,B_s)|\,\mathrm{d}s.$$

Private and routine martingale suprema tend to zero in L^p. The stopped drift, board, and count differences close under ordinary expectation and Gronwall. Hence the stopped error tends to zero for fixed H. Domination (3) gives uniform exponential moments, so the exit probability tends to zero uniformly as $H \to \infty$ and the full errors are uniformly integrable. Taking $N \to \infty$ first, then $H \to \infty$, proves expected uniform convergence. It gives no declared root-N constant after localization.

For tags, apply Lemma A.3 to the board convergence just proved, using the common proposal strip $[0, L]$ in place of $[0, \lambda_M]$. It gives (38) without a second tagged argument. □

7.3 The retained noise has a productive tagged effect

Use the early policy $\theta = \varepsilon \mathbf{1}_{[0,t_0)}\mathbf{1}_{\{C=\mathsf{A}\}}$ in the seeded region of Theorem 6.3. At every possible limit history, $a_t \geq a_0 e^{-\delta t}$ since the adopter inflow is nonnegative. Consequently the first-discovery hazard before t_0 is bounded below by $\lambda_M \varepsilon a_0 e^{-\delta t} > 0$. The common process has both a positive probability of no proposal on $[0, T]$ and a positive probability of exactly one accepted proposal in an early subinterval.

Let $H(Y) = \int_0^T e^{-\beta t} R X_t \,\mathrm{d}t$ for a limiting tag. On no proposals $m(\Pi^0) := \mathbb{E}[H(Y) \mid \mathcal{F}^0_T]$ equals the deterministic baseline reward. On exactly one early accepted proposal at time $t \leq t_0$, its value exceeds baseline by at least $e^{-\beta t} L_* m_* e^{-c_0 t}$: before the proposal the baseline still has an incomplete-unexposed fraction at least $m_* e^{-c_0 t}$. Exactly one proposal in an early subinterval and no others has positive probability, with acceptance probability at least $\lambda_M \varepsilon a_0 e^{-\delta t}/L > 0$. Thus m is nonconstant. Conditional independence yields

$$\mathrm{Cov}(H(Y_1), H(Y_2)) = \mathrm{Var}(m(\Pi^0)) > 0.$$

This establishes nontrivial unconditional dependence of reward path functionals, not merely randomness of an unused board. No assertion about the sign of every instantaneous state covariance is needed.

7.4 Scope of the convergence results

The preceding theorem is convergence under a fixed policy and hence, for this bounded payoff and Lipschitz feedback, convergence of that policy's welfare. Equilibrium values converge to the no-exploration values by (36), and unscaled equilibrium efforts converge uniformly to zero by (26). These are separate results. No convergence of Nu_i^{NE}, of optimal values, or of optimal finite-N controls to a limiting optimizer is claimed.

The secondary, formal limiting planner equation is

$$\begin{aligned} 0 = {} & \partial_t W + F(\mu,b)\cdot D_\mu W + [-\kappa b + w\eta\lambda_w a(\mu)]\partial_b W - \beta W + \langle \mu, g\rangle \\ & + a(\mu)\max_{0\leq u\leq \bar{u}}\{\lambda_M u[W(t,\mu,b+h) - W(t,\mu,b)] - c(u)\}, \qquad W(T) = 0. \end{aligned} \tag{39}$$

The disappearance of the private $1/N$ factor is already exact in the finite average-welfare HJB: marginal cost and event productivity both carry $1/N$. Classical differentiability, optimal-policy convergence, and a limiting control verification theorem for (39) remain separate obligations.

8 Creator credit: implementation and its boundary

8.1 Exact two-state implementation

At each discovery, attribute the success to one creator. Pay that creator Np and charge every player p, including the creator. If j creates, player i's net transfer is

$$T_i^j = p(N\mathbf{1}_{\{i=j\}} - 1), \qquad \sum_i T_i^j = 0.$$

This is exact event-by-event balance, not merely an expectation statement. If rivals use u and player i deviates to x in state zero, its discovery and transfer contribution to the Bellman equation is

$$\begin{aligned} &a_N x[D_i + (N-1)p] + a_N(N-1)u[D_i - p] \\ &\quad = a_N[x + (N-1)u]D_i + a_N(N-1)p(x-u). \end{aligned} \tag{40}$$

Thus

$$\beta V_{i,0} = \max_{0\le x\le \bar{u}} \{-c(x) + a_N x[D_i + (N-1)p] + a_N(N-1)u[D_i - p]\}, \qquad \beta V_{i,1} = v - dD_i.$$

The deviator does not change the transfer charged when a rival creates.

Proposition 8.1 (Balanced first-best credit in the benchmark)**.** *For either productivity normalization, either cost specification, and any cap, let D_{SP} be the corresponding two-state planner value gap. Setting $p = D_{\mathrm{SP}}$ implements the planner effort as a symmetric stationary Markov equilibrium of the transfer game. This is an implementation result, not a claim of uniqueness among all equilibria of that game.*

Proof. At equal efforts, each player's creator intensity is the same, and the expected instantaneous net transfer is zero. Its value therefore equals the planner's average value, giving $D_i = D_{\mathrm{SP}}$. By (40) the maximizing effort is

$$\mathrm{clip}_{[0,\bar{u}]}\left(\frac{a_N[D_{\mathrm{SP}} + (N-1)p] - \ell}{\gamma}\right) = \mathrm{clip}_{[0,\bar{u}]}\left(\frac{Na_N D_{\mathrm{SP}} - \ell}{\gamma}\right) = u_{\mathrm{SP}}.$$

The Bellman equations hold with the planner values; discounted jump compensation verifies the maximizer against all admissible deviations exactly as in Proposition 4.1. For each fixed N, bounded efforts and bounded event payments give integrable discounted transfers. The proof includes zero and cap-bound choices. At $N = 1$ the net transfer is zero. □

The result assumes risk neutrality, reliable success attribution, enforceable levies even on noncreators, commitment to future payment rules, and participation in the funding institution. Free exit before a levy, fake or misattributed artifacts, limited liability, or risk-bearing and verification costs change the game. Transfers cancel from welfare only when those additional costs are absent. The gross prize is order N, financing is order one per person per event, and the private marginal incentive is order one under $a_N = \lambda/N$. An order-one gross creator prize alone supplies only an order-$1/N$ direct marginal incentive.

8.2 Rare large payments and expectations

For the active normalized planner profile, the system discovery process has an N-independent law. Assume $T > 0$ and either an absent initial state, or a present initial state with $d > 0$, so $\mathbb{E}J_T > 0$. For a fixed horizon let C_i^N be player i's attributed success count. Conditional symmetry gives $\mathbb{E}C_i^N = \mathbb{E}J_T/N$ and $\mathbb{P}(C_i^N \geq 1) \leq \mathbb{E}J_T/N$. Thus the gross payment NpC_i^N tends to zero in probability but has expectation $p\mathbb{E}J_T > 0$. It is not uniformly integrable. With discounting, the same argument applies to $Np\int_0^T e^{-\beta t}\,\mathrm{d}C_i^N(t)$.

Under a common construction of the system events and uniformly assigned creator marks, the net tagged payment converges in probability to $-pJ_T$, although its finite-N expectation is zero. State-path weak convergence therefore does not preserve expected transfer payoffs. A continuum mechanism must retain compensators or a marked reward accounting with a separate expectation argument; uniform integrability of raw tagged payments is false in this example, not just an unverified technical condition. Finite-N compensator calculations suffice for the implementation theorem.

Even when expected transfer values can be bounded uniformly using their intensities, the no-transfer lemma cannot be reused: a creator's Hamiltonian contains the immediate $(N-1)p$ payment in addition to the bounded board-value difference.

8.3 A precise six-state criterion

Fix a bounded Borel symmetric target policy with common eligible effort $v(t,\mu,b)$, and let $p(t,\mu,b)$ be a bounded Borel per-event levy. Retain creator identity only as a contemporaneous payment mark. At the target, the instantaneous expected transfer to player i is

$$\lambda_M v p\,(\mathbf{1}_{\{C_i=\mathsf{A}\}} - a(\mu)). \tag{41}$$

This is generally nonzero. Let U_i^p be the actual continuation payoff under this policy and transfer rule. It is determined by the linear, verified evaluation equation

$$(\partial_t + \mathcal{L}_N^v - \beta)U_i^p + g(y_i) - \mathbf{1}_{\{C_i=\mathsf{A}\}}c(v) + \lambda_M v p(\mathbf{1}_{\{C_i=\mathsf{A}\}} - a) = 0, \qquad U_i^p(T) = 0,$$

interpreted mildly. For finite N, bounded transfers and intensity give this well-defined bounded potential.

Proposition 8.2 (Conditional implementation criterion)**.** *The target is a verified Markov equilibrium if, at every eligible state,*

$$v = \mathrm{clip}_{[0,\bar{u}]}\left(\frac{(\lambda_M/N)[\Delta_h U_i^p + (N-1)p] - \ell}{\gamma}\right). \tag{42}$$

For merely Borel rules, necessity holds almost everywhere in time along the target laws from each starting time and state (not only the given initial law); changes at a single time do not alter a payoff. For continuous rules and continuous evaluated values the condition is necessary pointwise at every preterminal state, by a short-time deviation from that state. If the target is an interior planner optimum, it requires $\Delta_h U_i^p + (N-1)p = N\Delta_h W_N^{\mathrm{SP}}$. Consequently the naive choice $p = \Delta_h W_N^{\mathrm{SP}}$ requires $\Delta_h U_i^p = \Delta_h W_N^{\mathrm{SP}}$ for every eligible player, which does not follow from budget balance or exchangeability.

Proof. For a unilateral deviation the direct event reward has exactly the algebra in (40), with $a_N = \lambda_M/N$, eligibility, and state-dependent p. Rivals retain their feedback rules and transfer evaluation is contained in U_i^p. Strict concavity and the bounded-potential verification identity prove

sufficiency. For Borel rules, the evaluation and comparison identities can be obtained directly by conditioning at an independent constant-rate proposal clock: use the baseline killed kernel up to the first proposal, then the accepted/rejected continuation. Iteration is justified by bounded payoffs and the Poisson tail. Thus this version does not assume continuity of a Borel potential for optional stopping. For necessity, mix the target action equally with the measurable clipped maximizer. Concavity gives a nonnegative residual, strictly positive wherever the criterion fails. Actual rare intensity is at least half target intensity; bounded-rate likelihoods make the target law absolutely continuous with respect to this deviating law. Positive target occupation of a strict residual would give a profitable deviation. Necessity follows from every restart; continuity strengthens it to pointwise necessity before T. Budget balance implies $N^{-1}\sum_i U_i^p = W_N^v$, an average identity that does not imply the individual jump identities. □

8.4 An exact counterexample to the naive lift

Take $N = 2$, rare-only mode, $\ell = 0$, and

$$B_0 = 0, \quad Y_1(0) = (1, \mathsf{A}), \quad Y_2(0) = (0, \mathsf{U}), \quad \delta = \nu = r = c_0 = 0, \quad R, \rho, \alpha, c_1, h, \lambda_M > 0.$$

Agent 1 remains a completed adopter forever. Agent 2 proceeds only $\mathsf{U} \to \mathsf{E} \to \mathsf{A}$ and then completes its task. Choose $T > 0$ small enough that $\lambda_M K_T/\gamma < \bar{u}$. The finite planner's feedback is then unbound by the cap. Let $W = W_2^{\mathrm{SP}}$ and propose the naive rule $p = \Delta_h W$ at its optimal profile $v = \lambda_M(\Delta_h W)_+/\gamma$.

Once agent 2 is exposed, the board cannot affect any further reward, so the planner sets all efforts to zero and $p = 0$. While agent 2 is unexposed, only agent 1 is eligible. Whenever $v > 0$, $p = \gamma v/\lambda_M$, and the target's cost and transfer rate for agent 1 cancel *exactly*:

$$-\frac{\gamma v^2}{2} + \frac{\lambda_M}{2} v(2-1)p = 0.$$

When $v = 0$, both terms vanish. Thus in every state of this invariant class $U_1^p(t,s) = K_{T-t}$, independently of the board, and $\Delta_h U_1^p = 0$. At an active target state, using this *target continuation value*, the effort-dependent Hamiltonian is

$$H(x) = \frac{\lambda_M p}{2} x - \frac{\gamma x^2}{2} = \frac{\gamma v x}{2} - \frac{\gamma x^2}{2}, \qquad H(v/2) - H(v) = \frac{\gamma v^2}{8} > 0.$$

This calculation is not a solution of the full dynamically optimal best-response problem. To establish its payoff implication, from (t,s) use $v(r, S_{r-})/2$ for $r \in [t, t+\tau]$ and revert to the target thereafter. Agent 1 remains eligible. The evaluated-potential comparison with transfers gives exactly

$$J_1^{\mathrm{dev}}(t,s) - U_1^p(t,s) = \mathbb{E}_{t,s}^{\mathrm{dev}} \int_t^{t+\tau} e^{-\beta(r-t)} \frac{\gamma v(r, S_{r-})^2}{8} \, \mathrm{d}r = \tau \frac{\gamma v(t,s)^2}{8} + o(\tau) > 0$$

for sufficiently small $0 < \tau < T - t$. The last equality follows from continuity of the planner feedback, stochastic continuity from the finite initial board, and bounded domination. More generally the same residual is nonnegative when the halved policy is used up to a stopping time. There is no gain from changing an action at one isolated time.

There is positive expected occupation of active states under the target from the displayed zero-board initialization. Otherwise the target effort and its artifact intensity would vanish $\mathrm{d}t\,\mathrm{d}\mathbb{P}$-almost everywhere, so its welfare could not exceed zero exploration, contradicting Theorem 6.3 with $a_* = m_* = 1/2$. Dividing $[0,T]$ into finitely many intervals shows that at least one interval

has positive expected integral of v^2. Halve the target on that interval, after following it up to the interval's start, then revert. The deviating and target path laws are equivalent on this finite interval: accepted rare intensities have ratio $1/2$ wherever nonzero (and are both zero elsewhere), and all other transition rules coincide. The finite-count likelihood has a strictly positive density. Thus the residual integral is strictly positive under the deviation too. This proves failure of implementation from the stated initialization, as well as failure of Markov perfection at active states.

The counterexample establishes a profitable short-time deviation to half the prescribed effort, not a complete best-response policy. It retains homogeneous costs, perfect attribution, exact budget balance and a verified planner. Existence of individualized funding satisfying (42) is not established here.

9 Related work and the contribution

The economic comparison builds on dynamic public-good and team-production models. Its specific contribution is to connect private production incentives to the limiting shared information environment, with a productive coordinated policy as the welfare comparison. The probability and verification arguments use established tools specialized to the present production institution.

Persistent fields and physical scaling. Persistent traces that mediate later behavior are established in stigmergy; Boldini et al. (2024) develop continuum and discrete descriptions of agents interacting through an evolving trace field. Our scalar board omits their spatial structure. Environmental-reservoir models likewise couple deposition, decay and pickup: Tien and Earn (2010) introduce a water compartment and study transmission thresholds, while Lanzas et al. (2020) discuss environmental transmission functions and their scaling. The loop gain in Appendix C is a model-specific specialization of that familiar structure, not a new reproduction-number principle. The physical deposition orders in Proposition 3.1 concern uncentered positive increments; they supply no centered diffusion or heavy-tail theorem.

Discovery, durability, and access. Bonatti and Hörner (2011, Sections II–III) already establish free riding and delay in team discovery with uncertainty and hidden effort. Their known-viability case (footnote 11) has individual effort of order $1/n$ without vanishing total effort. Declining individual effort alone is therefore not the contribution. Keller and Rady (2010, Sections 2–5) study genuine Bayesian experimentation through Poisson bandits; our known productivity and persistent usable board have no posterior learning.

Battaglini et al. (2014, Sections I–III) study a depreciating public stock with fixed aggregate resources W and individual endowments W/n. Their resource normalization differs from our fixed physical readiness cost and shrinking access probability. It supplies no empirical justification for uniform opportunity assignment. Battaglini and Palfrey (2024, Section 5, Theorems 4–5), a working paper revised in June 2026, show large-population efficiency or failure depending on how a participation threshold scales; their technology and private-information model differ from ours.

The broader connection between allocation of authority and private initiative is established, for example by Aghion and Tirole (1997). Proposition 4.2 is a narrower calculation about exogenous access probabilities; it does not solve their organizational choice problem. It demonstrates that a fixed opportunity budget alone is insufficient. Uniform dispersed access and pre-assignment readiness costs do the additional economic work.

Probability and verification tools. Hernández-Hernández and Ricalde-Guerrero (2025, Assumptions 1–2, Theorems 7 and 9) develop conditional McKean–Vlasov limits with common marked Poisson noise, using a product noise construction and regularity and moment assumptions. Our common-proposal coupling adapts established machinery. The integrated width of a thinning mismatch, Poisson domination, and finite-state martingale bounds are elementary; no verbatim embedding of the unbounded exposure model into their full theorem is asserted.

Neumann and Seifried (2026, Theorem 3.11) prove short-horizon existence for a coupled mean-field forward–backward system under Lipschitz reduced coefficients and aggregate-independent relocation. That published result solves a different problem from the finite-N backward construction here. The kernel semigroup perturbation theorem of Kühn and Kunze (2022, Hypothesis 3.1, Theorem 3.2) requires a strong Feller baseline. Our rare-disabled board has deterministic flow in rare-only mode and need not smooth Borel functions, so that theorem cannot be imported directly. Appendix B.1 proves the potential and interlacing identities under this model's actual clock construction. Banach contraction and reward-bound continuation are proof tools, not new general game-existence machinery.

Delarue and Foguen Tchuendom (2020, Section 6, Theorem 19) study randomized selection among deterministic mean-field equilibria. Their finite-player discussion analyzes first-order systems and explicitly distinguishes them from establishing Nash existence. This precedent motivates caution about broad novelty claims for incentives and limiting randomness; it is not an existence theorem for our game. Holmström (1982, Section 2) studies budget-balanced output sharing in teams. Attributed creator prizes use extra creator information and enforceable levies; they do not overturn an output-only sharing impossibility.

The connected result. The analysis supplies a finite-population equilibrium, uniform control of its path law and welfare, a productive fixed policy with an N-uniform gain, and strictly positive covariance of that policy's limiting tagged task rewards. The covariance statement ensures that the retained common history affects useful outcomes. Creator credit adds a boundary: budget balance alone does not transport a homogeneous benchmark implementation to heterogeneous private states. These are model-specific results, not claims of a general mechanism or of novelty from combining public-good and common-noise ideas.

10 Numerical methods and evidence

The calculations use a two-state policy evaluator and a six-state controlled piecewise deterministic simulator. A fixed configuration and independently seeded run records underlie the tables and figure.

The benchmark is checked against direct 2×2 linear policy evaluation, bounded scalar best-response optimization, planner policy optimization, occupancy ODE integration, and the convolution formula for first discovery from state one. The tests include $N = 1$, zero productivity and benefits, $d = 0$, positive linear costs, fixed productivity, and binding effort caps. They verify the proposed credit rule against a deviating policy rather than only substituting the same equilibrium formula twice.

The controlled simulator aggregates the 14 unchanged individual transition rates, retaining labels for state changes and creator attribution. Its thinning envelope is the current sum of individual and routine rates plus the constant rare bound $\lambda_M \bar{u}$. The first part decreases along the intervening board flow; the last part remains valid when effort rises with time or as the board decays. The candidate-time acceptance rate uses the decayed board and left-limit private state. Tests explicitly

start a time-increasing policy at zero current intensity, a case in which reusing the old total rate would fail. Generator tests verify recovery of the original mechanism at $u = 1$ and that $u = 0$ disables only rare production. Other checks cover the seed obstruction, simplex and board invariance, creator indices, and exact transfer balance.

Rewards are integrated exactly between jumps, including discounting. Constant early effort has exact integrated effort, cost, and hazard formulas; the two smooth diagnostic feedbacks use adaptive quadrature. Initial label permutation and dynamic events use independent child streams of the recorded SeedSequence. No board grid, Euler time step, or board truncation is used by the finite simulator. The fixed-policy limit uses exact Poisson proposal times and numerical ODE solutions between them, split at the policy's deterministic stopping time.

Table 3: Normalized quadratic benchmark, $\lambda = \beta = d = v = \gamma = 1$, $\ell = 0$ and $\bar{u} = 2$.

N	Nash effort	Planner effort	Nash value at 0	Nash mean wait
1	0.449490	0.449490	0.101021	2.2247
10	0.048866	0.449490	0.022685	20.4642
100	0.004988	0.449490	0.002475	200.4963
1000	0.000500	0.449490	0.000250	2000.4996

Table 4: Fixed coordinated early effort: independent-history means and 95% intervals.

N	Welfare above baseline	Mean artifacts	Integrated intensity
60	0.1147 [0.0469, 0.1825]	0.1125	0.1158
240	0.1947 [0.1184, 0.2710]	0.1750	0.1253
960	0.1240 [0.0618, 0.1863]	0.1062	0.1186
Limit	0.0998 [0.0442, 0.1554]	0.0750	0.1174

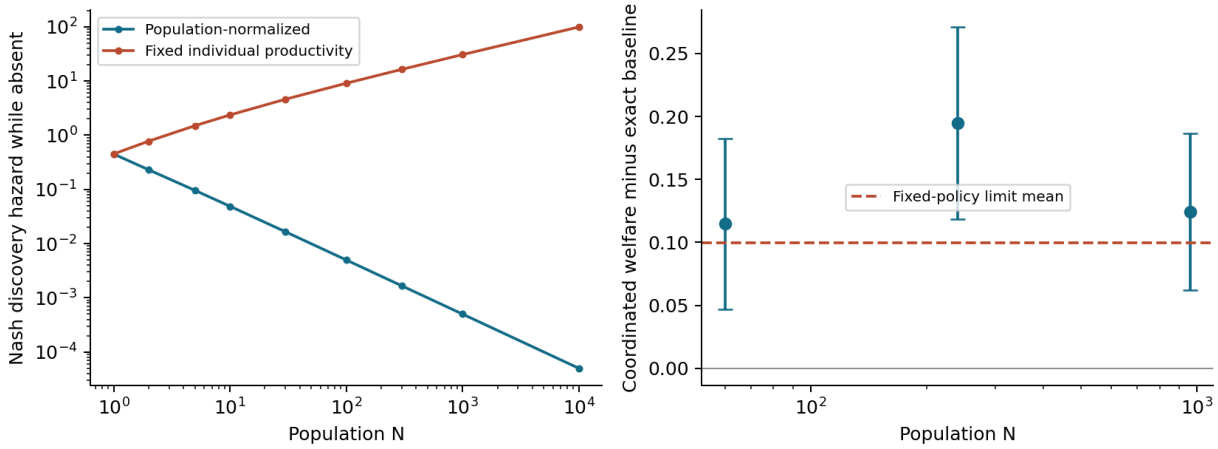


Figure 1: Left: the two-state equilibrium hazard under the two productivity normalizations. Right: the six-state fixed coordinated policy's welfare above its exact no-exploration baseline. Error bars use independent common histories.

The full-model experiment has $T = 6$, $\beta = 0.1$, $R = 1$, $\gamma = 0.1$, $\ell = 0$, $\bar{u} = 1$, and eligible effort 0.25 until time 1.25. Initial counts are 20% adopters and 80% incomplete-unexposed, all tasks initially

incomplete. Parameters are $\lambda_M = 2$, $h = 2$, $\rho = \alpha = 2$, $\kappa = 0.4$, $\nu = 0.15$, $\delta = r = 0.2$, $c_0 = 0.1$, $c_1 = 1.2$. The routine channel is off. Each cell uses 160 independent histories at $N = 60, 240, 960$; the limit has 160 independent histories. The seed root is 20260909, and complete seed formulas and per-run seeds are saved.

The exact no-exploration expected welfare is approximately 1.15065544. The separately computed first-order coefficients are approximately 0.576314, 0.576926, and 0.577079 at the three population sizes. These numerical coefficients use (50) and ODE/quadrature calculations, not simulation. The explicit theorem witness instead gives the deliberately conservative $A_* \simeq 9.61\,10^{-4}$, a certified effort of about $3.70\,10^{-7}$ and a certified gap of about $1.78\,10^{-10}$. The illustrated effort 0.25 is not certified by this coarse constant. Its larger welfare improvement is numerical evidence; the mathematical theorem certifies the existence of some fixed positive improvement.

The saved experiments also include zero effort and an imposed $0.25/N$ early-effort diagnostic. The latter is *not* a computed Nash equilibrium. Conditional hazards, realized counts, integrated intensities, task reward, average cost, and baseline-adjusted welfare are recorded separately. Student intervals use histories as independent replications; probabilities use Wilson intervals. They are finite-sample diagnostics with only 160 histories, not finely resolved estimates of extremely rare events. The artifact counts and integrated hazards are recorded independently, permitting a compensator check. Dependence of limiting tagged rewards is proved in Section 7; the sample variance of conditional mean reward is only an illustrative numerical diagnostic.

For this release, the numerical test suite was run afresh, deterministic coefficients and the witness constants were recomputed, and saved histories were checked against their reported aggregates. The displayed 1,440 finite and 160 limiting histories were retained, not resimulated. Numerical tests check implementation and numerical identities; they do not prove equilibrium existence or convergence. No numerical six-state Nash solve was performed.

The local editable release includes the complete LaTeX, bibliography, figure, tables and a build script. The public posting contains the paper and accompanying summary PDFs. Simulation code and run records remain in the local research project and are not part of this public release. The full parameter values, policy, replication counts and numerical methods are specified above; independent reproduction of the simulations from a public code package remains unavailable with this version.

11 Interpretation and next question

With exogenous production, physical deposition scaling separates a deterministic routine field from state-thinned common artifact jumps. The hybrid limit retains both, and the corresponding ordinary or conditional tagged product law makes the dependence distinction precise.

Under uniformly allocated, nontransferable opportunities and readiness paid before assignment, private incentives suppress the rare information channel as the population grows. The finite-population equilibrium and coupling bounds make this conclusion explicit. In the seeded, zero-board, rare-only productive region, a specified coordinated policy yields a positive per-capita gain and retains common jumps that induce unconditional dependence in tagged rewards.

The assumptions delimit the comparison. Equilibrium uniqueness is restricted to the bounded jointly continuous mild-HJB class. The welfare positivity result does not cover arbitrary initial boards or hybrid backgrounds. Fixed-policy convergence does not imply convergence of optimal values or controls, and the general controlled hybrid argument is qualitative. The physical deposition classification assumes an order-one eligible share; the strategic productivity-scaling classification and first-best creator-credit implementation belong to the two-state benchmark. In the six-state counterexample, a positive-duration half-effort deviation followed by reversion disproves the naive

implementation; a complete best-response policy and a general heterogeneous-state mechanism are not established.

The economic limitation is the access institution. Fixed capacity by itself does not force collapse, as designated access in the benchmark demonstrates. The paper supplies no evidence that uniform non-reassignable access and pre-assignment readiness describe a particular AI-agent system. Productivity is known, states are observed, and the scalar board omits content, permissions, and artifact-specific exposure. No Bayesian learning, endogenous hierarchy, efficient specialization, or empirical causal claim follows from the model.

A bounded next question is how reassignment of unused opportunities changes the comparison while holding the physical opportunity rate and task transitions fixed. Specifying that alternative allocation rule and the timing of readiness costs would test the economic premise directly. The present release records the conditional comparison at this stage of the research.

A Shared probability estimates and tagged paths

This appendix supplies the probability foundation for exogenous production and the artifact-disabled game. It also records the moment and tagged-coupling ingredients used in the controlled-policy comparison. All statements use the common model of Section 2; no predecessor result is invoked.

A.1 Construction, moments, and martingale decomposition

Write $M^{N,\mu}$ for the empirical martingale and $M^{N,w}$ for the compensated routine-board martingale (zero if the routine channel is absent). Independent private transition PRMs, a rate-$N\lambda_w$ routine proposal clock with independent uniform label marks, and the common artifact PRM give an explicit construction. A routine proposal is accepted only if its marked label is an adopter. In frozen mode an artifact proposal below height $m\lambda_M a(\mu^N)$ is accepted. For a symmetric game policy, replace this height by its predictable aggregate effort intensity. The creator can be sampled from the eligible intensity weights if attribution is needed; it has no state effect in the nontransfer models.

The board bound (3) makes the number of deposits finite on every bounded horizon. Private rates then have finite integrals. Successive event construction gives a pathwise solution for the driving-clock description and uniqueness in law. Between limit artifact proposals, the vector field $F(\mu, b)$ is locally Lipschitz, preserves the simplex, and has a nonnegative board with bounded deposition drift. This proves the corresponding limit's existence and uniqueness, recursively over its finitely many proposals.

Lemma A.1 (Uniform board and martingale bounds)**.** *Suppose $B_0^N \le K$ and the artifact intensity is bounded by a constant L_0 independent of N. For every fixed T, the board has uniformly bounded polynomial moments of all orders and exponential moments at every fixed positive exponent. For $p > 2$,*

$$\left\| \sup_{t\le T} \|M_t^{N,\mu}\|_1 \right\|_{L^p} + \left\| \sup_{t\le T} |M_t^{N,w}| \right\|_{L^p} \le C_T N^{-1/2}. \tag{43}$$

Proof. For $q > 0$, routine domination gives

$$\mathbb{E}e^{q\eta P_w(N\lambda_w T)/N} = \exp\{N\lambda_w T(e^{q\eta/N} - 1)\},$$

uniformly bounded in N; the rare proposal count has a fixed Poisson law and exponential moments at all finite exponents. Together with the initial bound, this proves the board claims. Random time

change gives

$$\mu_t^N = \mu_0^N + \int_0^t \mu_s^N Q(B_s^N)\,\mathrm{d}s + M_t^{N,\mu}, \tag{44}$$

$$B_t^N = B_0^N + \int_0^t [-\kappa B_s^N + w\eta\lambda_w a(\mu_s^N)]\,\mathrm{d}s + M_t^{N,w} + hJ_t^N. \tag{45}$$

The empirical coordinate jump sizes are at most $1/N$, and each predictable bracket is bounded by $CN^{-1}\int_0^T(1+B_s^N)\,\mathrm{d}s$. The jump Burkholder–Davis–Gundy bound for its pth supremum moment consists of the $p/2$ bracket moment and a term at most $CN^{1-p}\mathbb{E}\int_0^T(1+B_s^N)\,\mathrm{d}s$. The first has order $N^{-p/2}$ by the board moments, and $N^{1-p} \le N^{-p/2}$ for $p \ge 2$. Sum over the six coordinates. The routine martingale has jump size η/N and bracket

$$\langle M^{N,w}\rangle_t = \frac{w\eta^2\lambda_w}{N}\int_0^t a(\mu_s^N)\,\mathrm{d}s;$$

the same bound applies. This proves (43) without conditioning on a random board path as though it were independent of private clocks. □

A.2 A pathwise stability estimate for the common state dynamics

The generator structure gives a useful alternative to an exponential Gronwall factor involving the random board. For row vectors, define the operator norm induced by $\|\cdot\|_1$. Then

$$\|Q(b)\|_{1\to1} \le C(1+b), \qquad \|\mu[Q(b)-Q(\widetilde{b})]\|_1 \le 2\rho|b-\widetilde{b}| \quad (\mu\in\mathcal{P}(S)).$$

Only the exposure entries depend on b. A time-dependent finite-state Markov transition matrix is a contraction on signed row vectors in this norm.

Lemma A.2 (Empirical stability with a random board)**.** *Let μ^N satisfy (44) and let $\dot\mu = \mu Q(B)$, under a joint construction. Put $M_*^N = \sup_{s\le T}\|M_s^{N,\mu}\|_1$. Pathwise, for $t \le T$,*

$$\sup_{s\le t}\|\mu_s^N - \mu_s\|_1 \le \|\mu_0^N-\mu_0\|_1 + M_*^N\left[1 + C\int_0^T(1+B_s^N)\,\mathrm{d}s\right] + 2\rho\int_0^t |B_s^N - B_s|\,\mathrm{d}s. \tag{46}$$

Under Lemma A.1, the expectation of the martingale term on the right is $O(N^{-1/2})$.

Proof. Set $e_t = \mu_t^N - \mu_t$ and $h_t = e_t - M_t^{N,\mu}$. The jumps cancel, so h is absolutely continuous and, almost everywhere,

$$h_t' = h_t Q(B_t^N) + M_t^{N,\mu}Q(B_t^N) + \mu_t[Q(B_t^N) - Q(B_t)].$$

For each realized locally bounded board path, let $U(s,t)$ be the transition matrix generated by $Q(B_r^N)$ on $[s,t]$. Its entries are nonnegative and its rows sum to one, so $\|zU(s,t)\|_1 \le \|z\|_1$ for every signed row vector z. Ordinary variation of constants for the displayed pathwise ODE therefore bounds $\|h_t\|_1$ by its initial norm plus the integrals of the two forcing norms. Adding $M_t^{N,\mu}$ gives (46). No stochastic integral against the possibly future-dependent $U(s,t)$ is used. Hölder's inequality with p and $p/(p-1)$, the board moments, and (43) bound the expectation of the product term by $C_T N^{-1/2}$, irrespective of dependence between the board and martingale. □

A.3 Quantitative weak, rare, and frozen hybrid limits

Proof of Theorems 3.2, 3.3, and 3.4. Use the same artifact PRM for the finite and limiting frozen systems, with $w, m \in \{0,1\}$. Let K_t^N count proposals accepted by exactly one system. Predictable thinning gives

$$\mathbb{E}K_t^N = m\lambda_M \mathbb{E}\int_0^t |a(\mu_{s-}^N) - a(\mu_{s-})|\,\mathrm{d}s \le m\lambda_M \int_0^t \mathbb{E}\|\mu_s^N - \mu_s\|_1\,\mathrm{d}s. \tag{47}$$

All proposals, including rejected ones, are defined on the same strip. Hence $\sup_{s\le t}|J_s^N - J_s| \le K_t^N$. Subtract (45) from its limit equation to obtain

$$\begin{aligned}\sup_{s\le t}|B_s^N - B_s| \le |B_0^N - b_0| + \sup_{s\le T}|M_s^{N,w}| + hK_t^N \\ + \int_0^t [\kappa|B_s^N - B_s| + w\eta\lambda_w\|\mu_s^N - \mu_s\|_1]\,\mathrm{d}s.\end{aligned} \tag{48}$$

Combine expectations in (46)–(48). If

$$D_N(t) = \mathbb{E}\sup_{s\le t}(\|\mu_s^N - \mu_s\|_1 + |B_s^N - B_s| + |J_s^N - J_s|),$$

then Lemmas A.1–A.2 give

$$D_N(t) \le C_T(\varepsilon_{N,p} + N^{-1/2}) + C\int_0^t D_N(s)\,\mathrm{d}s.$$

Ordinary deterministic Gronwall proves the claimed rate for all three modes. For $m = 0$, both counts are zero and the limit is deterministic; $w = 0$ then gives the artifact-disabled, routine-disabled baseline as well. For $m = 1$, the recursive limit construction in this appendix gives the rare or hybrid common-jump process. Markov's inequality yields uniform convergence in probability under the coupling and consequently the stated product J_1 convergence. Lemma A.3 below gives the tagged conclusions.

In particular the hybrid estimate does not assume that the finite routine board is bounded by a deterministic constant conditional only on the rare proposal count. Its random size is controlled by the moment product in Lemma A.2. All constants refer to fixed T, fixed parameters, and (4); no uniform-in-time or changing-parameter rate is asserted. □

A.4 One tagged coupling for deterministic and common environments

Lemma A.3 (Tagged path factorization)**.** *Suppose the finite boards and a limiting board satisfy* $\mathbb{E}\int_0^T |B_s^N - B_s|\,\mathrm{d}s \to 0$. *Let the limiting population solve* $\dot{\mu} = \mu Q(B)$ *with deterministic initial law* μ_0, *and assume exchangeable* μ_0*-chaotic initial arrays independent of future driving clocks. If* B *is determined by a primitive common PRM, with private limiting clocks independent of it, bounded Lipschitz tagged path-product tests factor conditionally in the* L^1 *sense of* (9). *For deterministic* B, *the limiting path law is an ordinary product.*

Proof. For each fixed k, finite-state chaos gives a coupling of the initial k tags to independent μ_0 samples whose mismatch probability tends to zero. Use the same independent private transition PRMs for each tag and its limiting copy. Before their first disagreement, all non-exposure intensities agree; the exposure-thinning mismatch width is at most $\rho|B_s^N - B_s|$. Let $E_{N,k}$ be the event that any of these k paired paths disagrees on $[0,T]$. Compensation and a union bound give

$$\mathbb{P}(E_{N,k}) \le \mathbb{P}(\text{initial tag mismatch}) + k\rho\mathbb{E}\int_0^T |B_s^N - B_s|\,\mathrm{d}s \longrightarrow 0.$$

The independent limiting initial states and private PRMs construct chains with conditional marginal μ by uniqueness of the forward equation. Given the common PRM, these chains are independent with conditional path law $\mathcal{P}^0$. For bounded tests, a telescoping product bound and conditional Jensen bound the L^1 conditional-product error by a fixed constant times the probability just displayed. This proves conditional factorization (in fact for bounded measurable tests under this coupling). If B is deterministic, the conditional law is nonrandom, giving ordinary path-space chaos. With random B, integrating over the common PRM gives a mixture of products and need not give an unconditional product. □

A.5 Deposition orders

Proof of Proposition 3.1. With an order-one eligible share, N possible sources at individual rate $\lambda_0 N^{-\zeta}$ have aggregate rate of order $N^{1-\zeta}$. Multiplying once and twice by the positive impact $\eta_0 N^{-\xi}$ gives drift order $N^{1-\zeta-\xi}$ and predictable quadratic-variation order $N^{1-\zeta-2\xi}$. On $\zeta+\xi=1$, the latter is $N^{-\xi}$: it vanishes for $\xi>0$, whereas $\xi=0$ leaves finite intensity and finite jumps. Off that line, the positive first-order contribution either vanishes or grows under the specified eligible-share hypothesis. These exponent calculations do not provide a Brownian limit, a limiting eligibility law, or an equilibrium selection argument. □

B Verification and supporting estimates

B.1 Potential verification under predictable interlacing

All semigroup integrals here are pointwise bounded kernel integrals. No Bochner integral or global strong continuity on C_b is required. On a terminal slab $[a,b]$, define

$$Q_\xi\psi(t)=e^{-\xi(b-t)}P^0_{b-t}\psi, \qquad R_\xi f(t)=\int_t^b e^{-\xi(r-t)}P^0_{r-t}f(r)\,\mathrm{d}r.$$

The semigroup property and bounded Fubini give $Q_\beta - Q_{\beta+L} = LR_{\beta+L}Q_\beta$ and $R_\beta - R_{\beta+L} = LR_{\beta+L}R_\beta$. Thus $V=Q_\beta\psi+R_\beta F$ solves $V=Q_{\beta+L}\psi+R_{\beta+L}(F+LV)$. Conversely, for fixed bounded F the latter linear Volterra equation has a unique bounded solution: its homogeneous difference satisfies $\|D(t)\|\le L\int_t^b\|D(r)\|\,\mathrm{d}r$ and vanishes by iteration. Taking $F=g-c(A[V])+\Lambda[V]\Delta_h V$ proves killed/un-killed equivalence even though this F was originally defined through V.

Lemma B.1 (Controlled potential identity)**.** *Let V,F be bounded jointly continuous functions satisfying $V=Q_\beta\psi+R_\beta F$. Retain baseline private and routine transitions, and add $s\mapsto s^h$ jumps with any predictable intensity $0\le\lambda_r\le L$. For every bounded stopping time $\sigma\in[t,b]$,*

$$V(t,s)=\mathbb{E}_{t,s}\left[e^{-\beta(\sigma-t)}V(\sigma,S_\sigma)+\int_t^\sigma e^{-\beta(r-t)}\{F(r,S_{r-})-\lambda_r\Delta_h V(r,S_{r-})\}\,\mathrm{d}r\right]. \tag{49}$$

The associated process is a true martingale for the admissible filtration.

Proof. Splitting the mild identity at deterministic r shows that under a baseline restart from (t,s),

$$Z^0_r=e^{-\beta(r-t)}V(r,S^0_r)+\int_t^r e^{-\beta(v-t)}F(v,S^0_v)\,\mathrm{d}v$$

is the conditional expectation of the bounded terminal variable $e^{-\beta(b-t)}\psi(S^0_b)+\int_t^b e^{-\beta(v-t)}F(v,S^0_v)\,\mathrm{d}v$. It is uniformly integrable and cadlag, by joint continuity of V and cadlag baseline paths. Private

and routine future increments remain independent of the past when the filtration includes unused past proposal marks and independent control randomization, so this martingale property persists in the enlarged admissible filtration.

For $L > 0$, take independent rate-L proposal times σ_k with independent uniform marks U_k. Accept at σ_k if $U_k \le \lambda_{\sigma_k}/L$, evaluating the intensity before revealing U_k. After each proposal restart the baseline from the actual post-proposal state, using fresh future private and routine increments. Random restart times and states are allowed by the baseline strong Markov property. Until the next proposal the path is baseline. Optional stopping of its bounded potential martingale at that independent exponential time, cut off at b, gives a zero conditional mean segment increment. Predictable deviations can depend on the whole past; they affect acceptance and do not change the law of this baseline segment before its next proposal.

Sum the segment martingales, initially stopping after m proposals. There are almost surely no simultaneous private/routine and rare proposals. An accepted proposal adds exactly $e^{-\beta(\sigma_k - t)}\Delta_h V(\sigma_k, S_{\sigma_k -})$; a rejected one adds zero. Hence

$$e^{-\beta(r-t)}V(r,S_r) + \int_t^r e^{-\beta(v-t)}F(v,S_{v-})\,\mathrm{d}v - \sum_{\sigma_k \le r}^{\text{accepted}} e^{-\beta(\sigma_k - t)}\Delta_h V(\sigma_k, S_{\sigma_k -})$$

is a stopped martingale. Its absolute value is bounded by a constant times $1 + P_L(b-t)$, where P_L is the rate-L proposal count. This integrable domination permits $m \to \infty$ in conditional expectations.

Predictable thinning gives compensator $\lambda_v\,\mathrm{d}v$. Subtract from the displayed jump sum its compensator

$$\int_t^r e^{-\beta(v-t)}\lambda_v \Delta_h V(v,S_{v-})\,\mathrm{d}v.$$

The result is a true martingale: the integrand is bounded by $2\|V\|_\infty$ and the expected count by $L(b-t)$. Adding that compensated martingale to the preceding one gives the claimed martingale. Its expression is bounded by

$$\|V\|_\infty + (b-t)(\|F\|_\infty + 2L\|V\|_\infty).$$

Optional stopping gives (49). For $L = 0$, the baseline identity suffices. □

Under a unilateral deviation, rivals retain $A_j[V](r, S_{r-})$, and

$$\lambda_r = \frac{\lambda_M}{N}\left(u_i(r) + \sum_{j \ne i} A_j[V](r, S_{r-})\right).$$

Rival actions and values are evaluated at the deviating state. The lemma with $\sigma = b$ gives the main proof's Hamiltonian residual and includes all future feedback responses. Applied to the rare-disabled welfare potential, it also proves (30) for a Borel gate: the gate enters only the bounded predictable actual intensity, not the continuous V or F.

B.2 A precise numerical formula for the one-artifact coefficient

In rare-only mode at zero board and zero effort, labeled chains are independent. Let $p_j(t,\cdot)$ be the distribution of a chain initially in state j, f_j the initial deterministic fraction in that state, and

$$d_y(t) = w_y(T-t,h) - w_y(T-t,0),$$

where $w_y(H, b)$ is the remaining discounted reward of a single chain with deterministic decaying board $be^{-\kappa r}$ and no further deposits. Then $\Delta_h W_N^0 = \sum_y \mu^N(y) d_y$, and writing $\bar{a} = \sum_j f_j p_j(\mathsf{A})$ and $\bar{d} = \sum_j f_j \sum_y p_j(y) d_y$ gives

$$\mathbb{E}^0[a(\mu_t^N)\Delta_h W_N^0] = \bar{a}\bar{d} + \frac{1}{N}\sum_j f_j \left[\sum_{y:C(y)=\mathsf{A}} p_j(y)d_y - p_j(\mathsf{A})\sum_y p_j(y)d_y\right]. \tag{50}$$

To derive it, expand the product of the two empirical averages. Distinct labels' terms factor; equal-label terms contribute exactly the displayed covariance. This retains the $1/N$ correction rather than replacing an expected product by a product of expectations.

The code computes $p_j(t)$ using the matrix exponential of Q_0. For w_y, it solves $P'(r) = P(r)Q_{be^{-\kappa r}}$, $P(0) = I$, together with $z'(r) = e^{-\beta r}P(r)g$, $z(0) = 0$, and returns $z(H)$. At $b = 0$ this is checked against the independent matrix-resolvent identity

$$z(H) = (Q_0 - \beta I)^{-1}(e^{(Q_0-\beta I)H} - I)g$$

when $\beta > 0$. Gaussian quadrature of (50) gives A_N; quadrature-order refinement is a numerical check, not a proof of positivity.

B.3 Regularity boundaries

The martingale estimates needed by the baseline and controlled limits are proved once in Lemma A.1. The rare-disabled population LLN is Theorem 3.2; no separate manuscript is needed for that estimate.

For the equilibrium proof, the semigroup argument operates on continuous bounded functions, not on a putative bounded classical generator. The potential's martingale identity supplies the needed extended-generator property. Local contraction alone does not prove global existence: verification and the uniform reward bound are the continuation step. Compact controls and strict concavity alone would not solve a general coupled game without the explicit maximizing selector in (22).

C Exposure/adoption specialization

The six individual states yield a seven-coordinate deterministic weak limit: six state shares plus the board. Its invariant state manifold is $\Delta(S) \times \mathbb{R}_+$, where $\Delta(S)$ is the probability simplex on S. Because communication rates do not depend on the private task coordinate, summing over that coordinate closes the communication subsystem. Let υ, e, a be the unexposed, exposed, and adopter shares, with $\upsilon = 1 - e - a$, and write b for the weak board limit. Then

$$\dot{e} = \rho b(1 - e - a) - (\alpha + \nu)e, \qquad \dot{a} = \alpha e - \delta a, \qquad \dot{b} = \eta\lambda_w a - \kappa b. \tag{51}$$

Here exposure is separated from first behavioral use. Exposed agents either adopt at rate α or forget at rate ν; adopters abandon at rate δ; adopters post at rate λ_w with yield η; and the board decays at rate κ.

Proposition C.1 (Communication threshold)**.** *Assume $\alpha, \delta, \kappa, \eta, \lambda_w, \rho > 0$, $\nu \geq 0$, and $c_0 + r > 0$. Define*

$$G_B = \frac{\rho\alpha\eta\lambda_w}{(\alpha + \nu)\delta\kappa}.$$

The uncoordinated equilibrium of the full weak system has $(e, a, b) = (0, 0, 0)$ *and* $(\mu(0, \mathsf{U}), \mu(1, \mathsf{U})) = (r, c_0)/(c_0 + r)$*. It is locally asymptotically stable on the probability-simplex tangent space for* $G_B < 1$ *and unstable for* $G_B > 1$*. For* $G_B > 1$*, the reduced system has the unique positive equilibrium*

$$v_* = \frac{1}{G_B}, \qquad e_* = \frac{1 - G_B^{-1}}{1 + \alpha/\delta}, \qquad a_* = \frac{\alpha}{\delta} e_*, \qquad b_* = \frac{\eta \lambda_w}{\kappa} a_*.$$

At $G_B = 1$ *the branches meet in a forward transcritical bifurcation. The reduced positive branch is locally asymptotically stable wherever feasible, and the seven-coordinate system, restricted to the invariant state manifold* $\Delta(S) \times \mathbb{R}_+$*, has local stability exchange in a neighborhood of the bifurcation point. No global stability statement is made.*

The aggregate characteristic polynomial at the uncoordinated equilibrium is

$$P(z) = (z + \alpha + \nu)(z + \delta)(z + \kappa) - \rho\alpha\eta\lambda_w.$$

The private-task directions are strictly stable under the stated assumptions, and Routh–Hurwitz gives the crossing described above. A null-vector calculation in the proof below verifies transversality and nondegeneracy.

The equation $\dot{a} = \alpha e - \delta a$ is not replicator dynamics. Adoption is an exposure-mediated flow, not payoff-based copying or replacement. The threshold is useful because it transparently combines posting, persistence, exposure, adoption, forgetting, and abandonment. Its form is nevertheless standard: environmental-reservoir epidemic models couple shedding into a decaying environment to later pickup and transition, and derive analogous loop gains (Tien and Earn, 2010; Lanzas et al., 2020). The interpretation of B as information rather than pathogen load changes the application, not the algebraic ancestry.

Proof. Write $r_e = \alpha + \nu$ and $L = \eta\lambda_w$. Linearization of (51) at zero gives

$$J_0 = \begin{pmatrix} -r_e & 0 & \rho \\ \alpha & -\delta & 0 \\ 0 & L & -\kappa \end{pmatrix},$$

whose characteristic polynomial is the one displayed in the main text. If $P(z) = z^3 + A_1 z^2 + A_2 z + A_3$, then

$$A_1 = r_e + \delta + \kappa > 0, \quad A_2 = r_e\delta + r_e\kappa + \delta\kappa > 0, \quad A_3 = r_e\delta\kappa(1 - G_B).$$

For $G_B < 1$, $A_1 A_2 > A_3 > 0$, so Routh–Hurwitz gives stability. At $G_B = 1$, zero is simple and the remaining quadratic is Hurwitz; for $G_B > 1$, $P(0) < 0$ and $P(z) \to \infty$, so a positive real root exists. On the full simplex, the remaining task-composition eigenvalues at the uncoordinated point are $-(c_0 + r)$, $-(c_0 + r + r_e)$, and $-(c_1 + r + \delta)$, all negative under the stated assumptions. Continuity preserves these transverse signs locally.

Solving the equilibrium equations gives the positive branch stated in the proposition. At a positive reduced equilibrium, putting $x = \rho b_* > 0$, the cubic coefficients satisfy

$$\begin{aligned} A_1 &= \delta + \kappa + r_e + x, \\ A_2 &= \alpha x + \delta\kappa + \delta r_e + \delta x + \kappa r_e + \kappa x, \\ A_3 &= x\kappa(\alpha + \delta). \end{aligned}$$

They are positive and $A_1 A_2 > A_3$, proving reduced local stability. Taking ρ as bifurcation parameter, at $\rho_c = r_e\delta\kappa/(\alpha L)$ choose right and left null vectors

$$v = \left(1, \frac{\alpha}{\delta}, \frac{L\alpha}{\delta\kappa}\right), \qquad w = \left(1, \frac{r_e}{\alpha}, \frac{\delta r_e}{\alpha L}\right).$$

Then

$$w^T Df_\rho(0)v = v_b \neq 0, \qquad w^T D^2 f(0)[v,v] = -2\rho_c v_b(v_e + v_a) \neq 0.$$

The standard transcritical nondegeneracy conditions hold, and the explicit branch has positive a_* derivative at ρ_c. Together with the stable transverse directions this proves the local full-system stability exchange.

□

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